2025-2026 – Econ 0039 – Advanced Macro

Lecture 2: Business Cycles

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Changes from version 1.0 are in red

Disclaimer

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- Define Business Cycle
- Explain how to measure it: time domain and frequency domain
- Show some of its properties (dynamics, comovements)
- Modeling: A dynamic reduced-form model and a microfounded general equilibrum model with supply and demand shocks

Plan of the Lecture

1. Definition
2. Time Domain
3. Frequency Domain
4. A Simple Dynamic Model of the Business cycle
5. Facts about Business Cycles
6. Demand and Supply Shocks in an “UV” model
7. Summary
8. References

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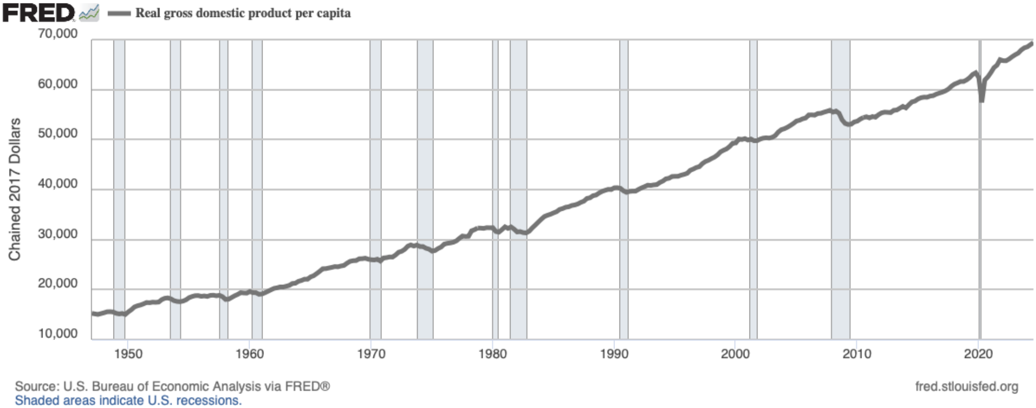
1. Definition

- Macroeconomics is about the determination of aggregate variables, as measured by national accounts (output, consumption, employment, inflation,...)
- Economists makes a distinction (at least at first pass) between the long run and the short run, between Growth and Business Cycle
- For the methodological part of that lecture, I will consider the U.S. of A. as the running example.
- Remark: I will not consider infra annual seasonal fluctuations (like ice-cream sales in winter and summer). All the series I will be using are *seasonally adjusted*.

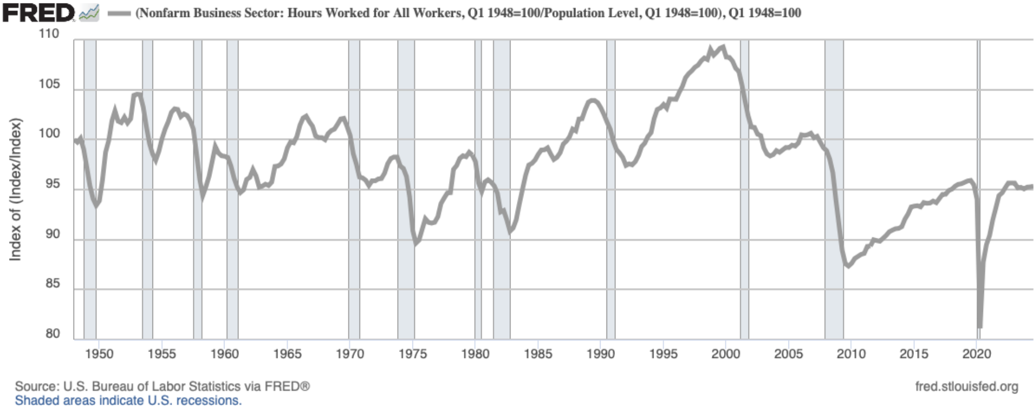
1. Definition

Figure 1: A Trending Series: US log Real GDP per capita



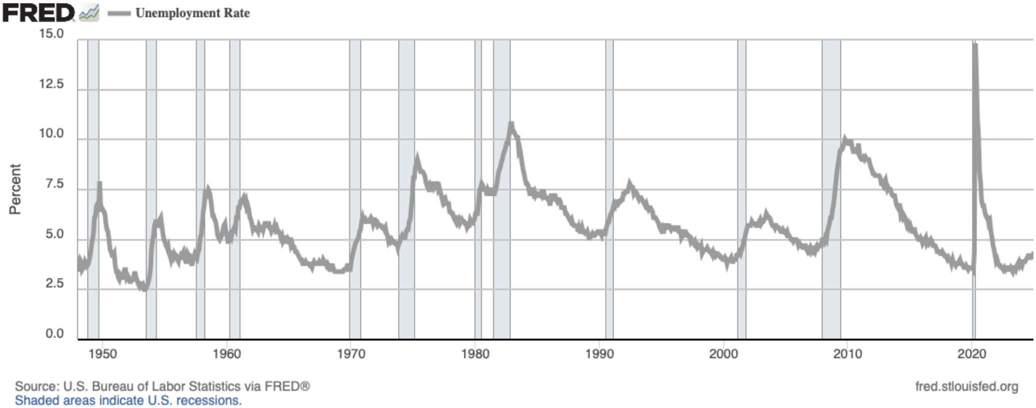
1. Definition

Figure 2: A Non-Trending Series: US Hours per capita



1. Definition

Figure 3: Another Non-Trending Series: US Unemployment rate



1. Definition

- Burns & Mitchell [1946] "Measuring Business Cycles", National Bureau of Economic Research:

  "Business cycles are a type of fluctuation found in the aggregate economic activity of nations that organize their work mainly in business enterprises: a cycle consists of expansions occurring at about the same time in many economic activities, followed by similarly general recessions, contractions, and revivals which merge into the expansion phase of the next cycle."

- To identify cycles, Burns & Mitchell assume that they are no shorter than 6 quarters, and found a maximum length of 32 quarters (1885 to 1931).

Figure 4: Reproduced from BURNS & MITCHELL [1946], *Measuring Business Cycles*

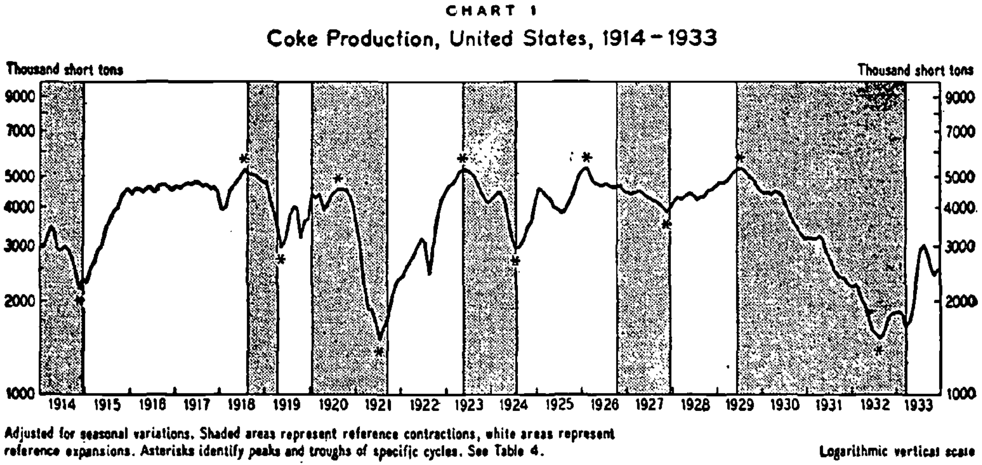
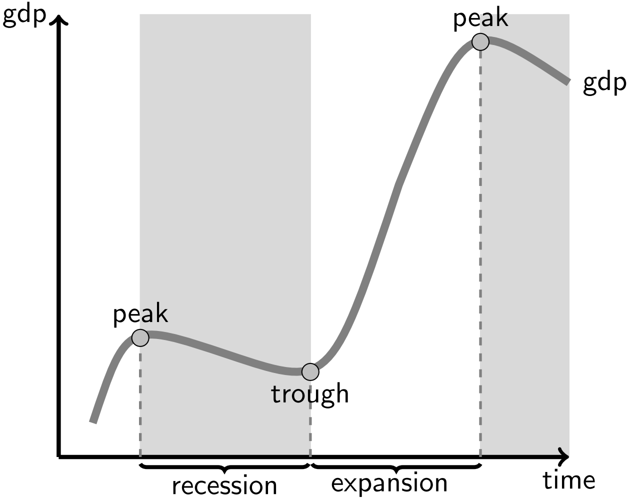
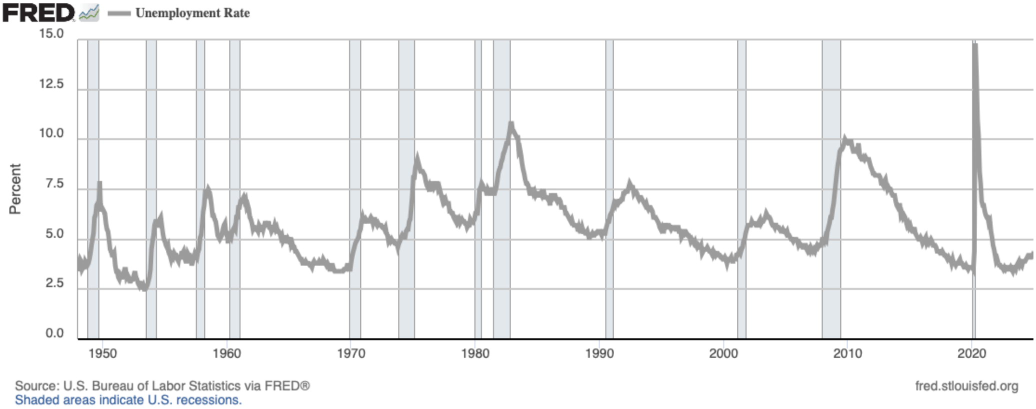


Figure 5: A Business Cycle



1. Definition

Figure 6: U.S. Business Cycles, as identified by the NBER's Business Cycle Dating Committee



1. Definition

Table 1: Recent U.S. Business Cycles, as identified by the NBER's Business Cycle Dating Committee

Business Cycle Reference Dates						Contraction	Expansion	Cycle	
Peak Month	Peak Year	Peak Quarter	Trough Month	Trough Year	Trough Quarter	Peak to Trough (Months)	Previous Trough to this Peak (Months)	Trough from Previous Trough (Months)	Peak from Previous Peak (Months)
April	1960	2	February	1961	1	10	24	34	32
December	1969	4	November	1970	4	11	106	117	116
November	1973	4	March	1975	1	16	36	52	47
January	1980	1	July	1980	3	6	58	64	74
July	1981	3	November	1982	4	16	12	28	18
July	1990	3	March	1991	1	8	92	100	108
March	2001	1	November	2001	4	8	120	128	128
December	2007	4	June	2009	2	18	73	91	81
February	2020	(Peak 2019Q4)	April	2020	2	2	128	130	146

- How does the NBER establish this chronology?
- Here is the "Statement of the NBER Business Cycle Dating Committee on the Determination of the Dates of Turning Points in the U.S. Economy".

"The NBER's Business Cycle Dating Committee maintains a chronology of the U.S. business cycle. The chronology comprises alternating dates of peaks and troughs in economic activity. A recession is a period between a peak and a trough, and an expansion is a period between a trough and a peak. During a recession, a significant decline in economic activity spreads across the economy and can last from a few months to more than a year. Similarly, during an expansion, economic activity rises substantially, spreads across the economy, and usually lasts for several years."

1. Definition

"In both recessions and expansions, brief reversals in economic activity may occur – a recession may include a short period of expansion followed by further decline; an expansion may include a short period of contraction followed by further growth. The Committee applies its judgment based on the above definitions of recessions and expansions and has no fixed rule to determine whether a contraction is only a short interruption of an expansion, or an expansion is only a short interruption of a contraction . The most recent example of such a judgment that was less than obvious was in 1980-1982, when the Committee determined that the contraction that began in 1981 was not a continuation of the one that began in 1980, but rather a separate full recession."

1. Definition

- Is there a way to translate this into some statistical procedure?
- Two difficulties (at least)
  1. For growing series (GDP per capita), separate trend from cycle
  2. But even if no obvious trend (Hours worked per capita), separate long movements (e.g. demographic) from business cycles ones
- We first look at ways to extract the cycle *in the time domain*
- Then we will look at another way of looking at times series, *in the frequency domain*

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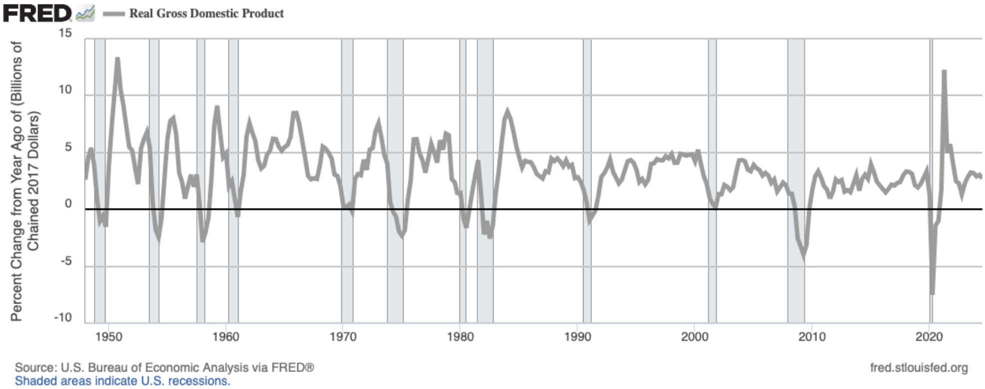
- Any time series $y_t = \log Y_t$ can be decomposed into a "Trend" and a "Cycle" such that
$$y_t = y_t^T + y_t^C$$
- Problem: How to define/identify each component?
- Several ways of approaching the problem
- Actually: Infinite number of decomposition of a non-stationary process into a cycle and a trend
- Let us see some "intuitive" definition of those decompositions

- Take the growth rate of the US Real GDP: $y_t^C = y_t - y_{t-1}$
- *(Reminder: the difference of logs is approximatively the growth rate)*
- Note: the cycle is very volatile (a lot of ups and downs) compared to the NBER chronology.

Figure 7: US Growth Cycles (Real GDP)



- Deviation from linear trend
- The trend is obtained from linear regression

$$y_t = \widehat{\alpha} + \widehat{\beta} t + \widehat{u}_t$$

- The “hats” show that the coefficients are *estimated* coefficients.
- Cycle: $y_t^C = y_t - (\widehat{\alpha} + \widehat{\beta} t)$
- Expansion: Output above the trend

Figure 8: US real GDP and Linear Trend

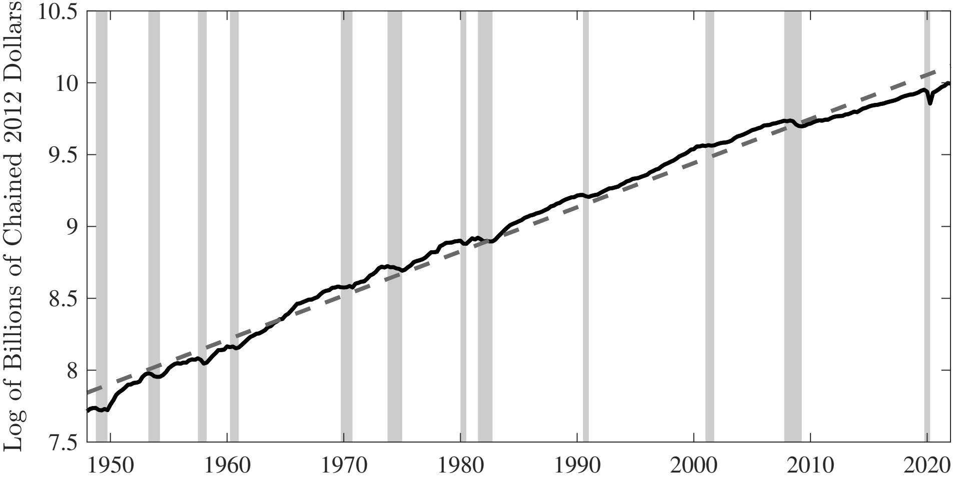
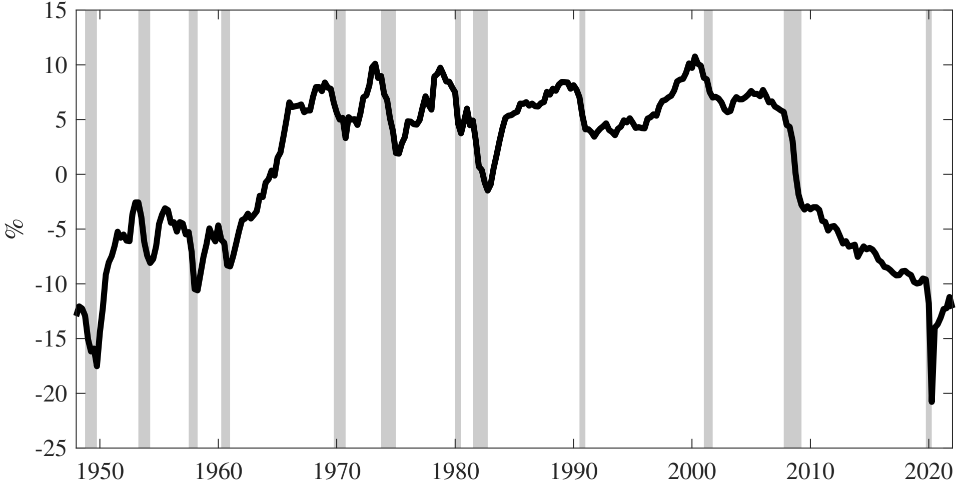


Figure 9: US real GDP Trend Cycles



- Note: the cycle can be large and very persistent.
- Some longer run phenomena are not eliminated (for example the productivity slowdown of the 1970's)

- Define the Output Gap as

$$\textit{Output Gap} = \text{-}(\textit{Actual Output minus Potential Output})$$

- Expansion: Actual Output $\geqslant$ Potential Output
- Actual output: easy to observe
- Note: How to identify potential output? (full utilization?, efficient?)
- Example:
  1. estimate $y_t = \alpha \times u_t + \text{other controls} + \varepsilon_t$,
  2. define potential output as $y_t^P = \widehat{\alpha} \times 0\% + \text{other controls} + \widehat{\varepsilon}_t$.
  3. Instead of $u = 0\%$, one might choose $u = u^n$ where u^n is the *natural* rate of unemployment (the Oecd chooses the NAIRU (Non Accelerating Inflation Rate of Unemployment))
- Cycle is then $y_t - y_t^P$

- This is an over simplified description of the method used for example by the U.S. Congressional Budget Office.

Figure 10: US Actual Real GDP and CBO Potential Output

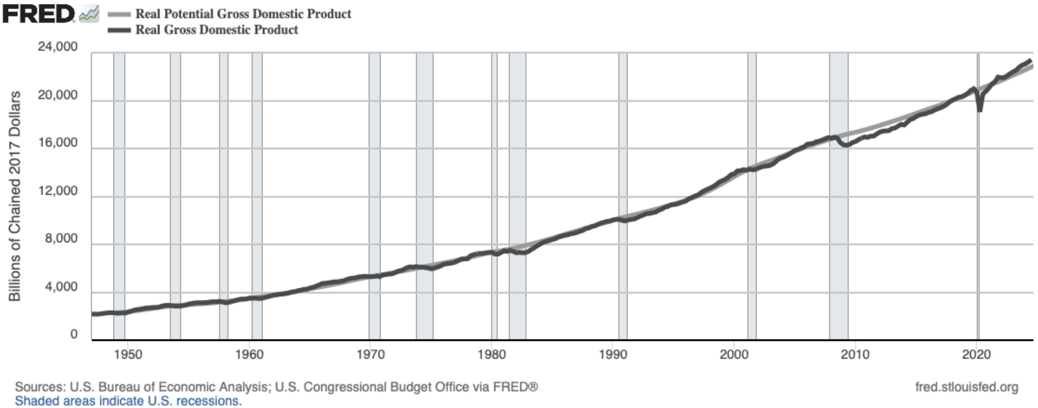
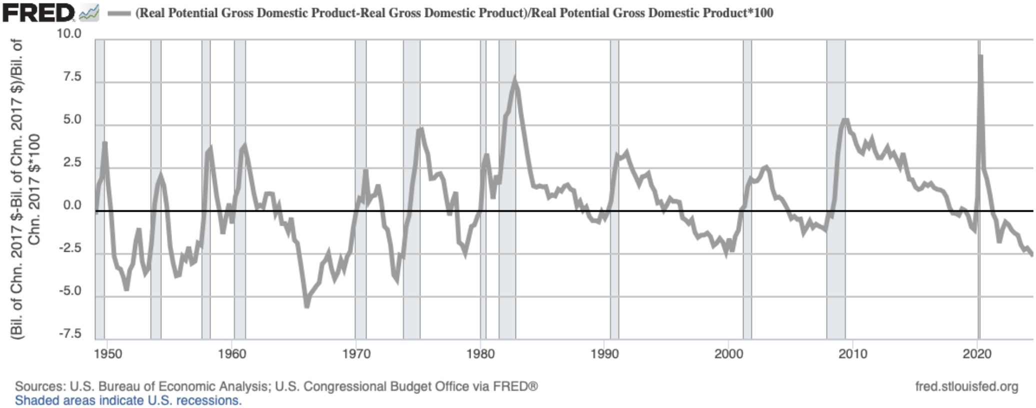


Figure 11: US CBO Output Gap (in % of actual GDP)



- This tracks pretty well the NBER expansions and recessions
- But a pretty complicated method that relies on a lot of assumptions. See Shackleton [2018], "Estimating and Projecting Potential Output Using CBO's Forecasting Growth Model".

- Very popular in the macro literature
- In the time domain, the idea is to remove a trend which is smooth, but not linear
- The trend y_t^T is the Argmin of:

$$\sum_{t=1}^{T}(y_t - y_t^T)^2 + \lambda \sum_{t=1}^{T-1}\left((y_{t+1}^T - y_t^T) - (y_t^T - y_{t-1}^T)\right)^2$$

× if $\lambda = +\infty$, y_t^T is a straight line.
× if $\lambda = 0$, then $y_t^T = y_t \ \forall t$.

- With $\lambda = 1600$ on quarterly data , it is believed to produce cycles not too far from those identified by the NBER.
- Literature has focused on that value.
- Why is that important? Because new theories tend to be evaluated in the following way:
  - × write the model, simulate, filter with HP filter (1600)
  - × compare to data filtered in the same way.
- What is filtered out is therefore not to be explained.

Figure 12: US real GDP and Hodrick-Prescott Trend

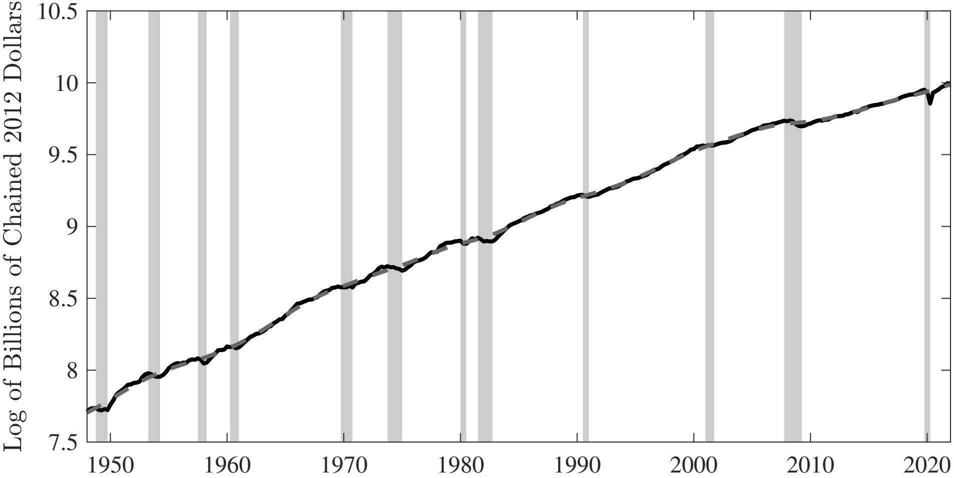
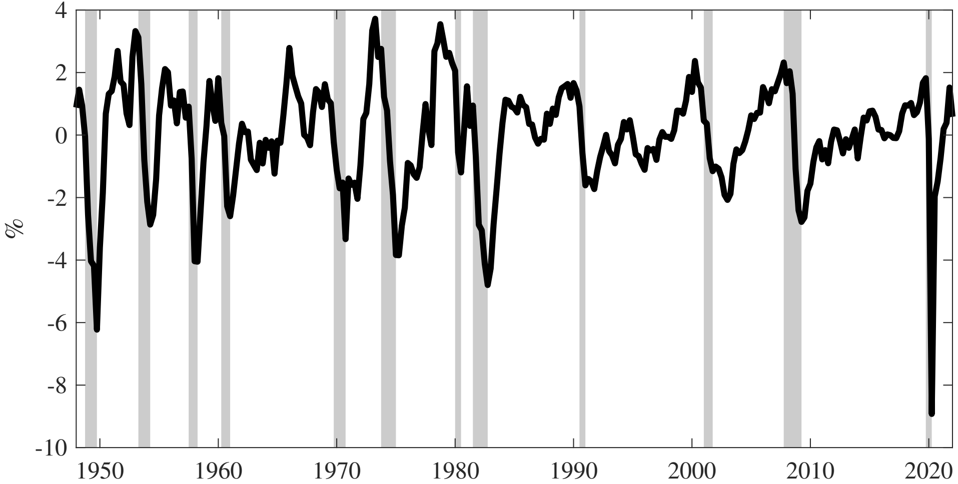


Figure 13: US Real GDP Hodrick-Prescott Cycle



- As I said, very popular in macro.
- Note that it keeps a lot of short run fluctuations that are not related with the NBER expansions and recessions.

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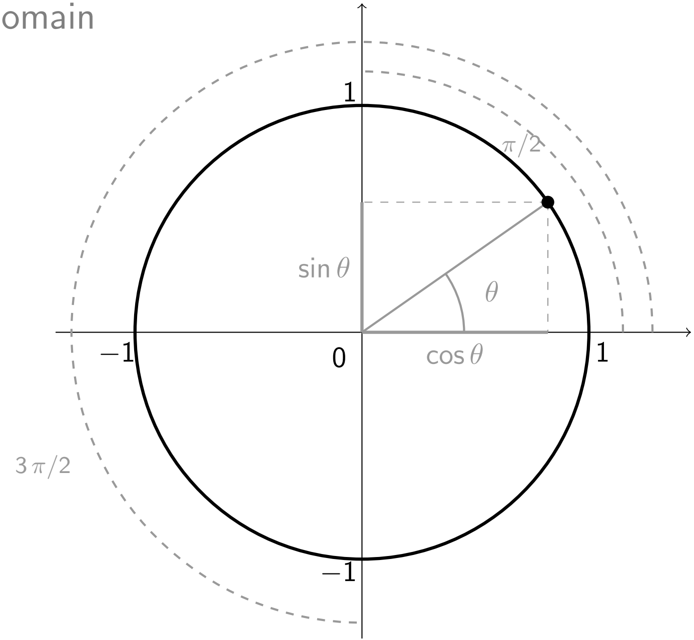
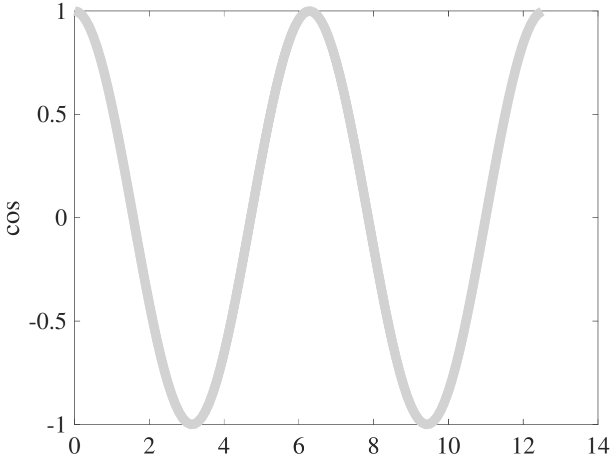
- When a series is stationary, one can express it as a sum of periodic functions, for example sine and cosine waves of various frequencies.
  - × Formally, the series needs to have constant first and second moments (mean, variance, autocorrelations)
  - × This result is known as *Fourier transform* of a function of time.
  - × Economic processes can therefore be analysed in the *time domain* (as we have done before) or in the *frequency domain* (as we will do now)

Trigonometry 101



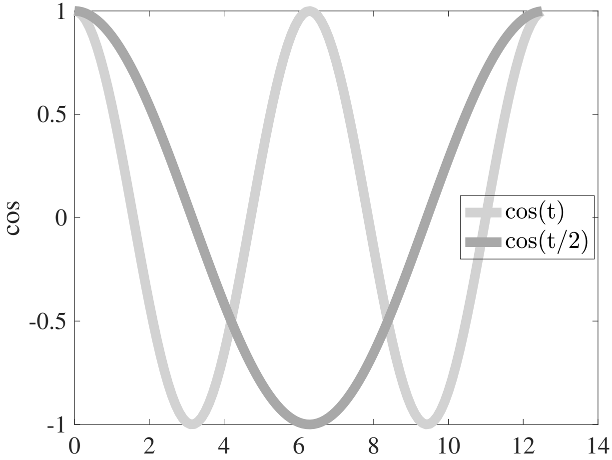
- A typical periodic function is $\cos(\omega t)$, with period (the time it takes to reproduce itself) $2\pi/\omega$.
  - × Knowing that period of $\cos(t)$ is 2π, for a given t_1, what is the t_2 such that $\cos(\omega t_2) = \cos(\omega t_1)$? It must be that $\omega t_2 = \omega t_1 + 2\pi$.
  - × The solution is $t_2 - t_1 = 2\pi/\omega$.
- $\frac{\omega}{2\pi}$ is the *frequency* of oscillation (number of cycles per unit of time)

Figure 14: Cosine wave with ω=1 ($x_t = \cos(t)$)



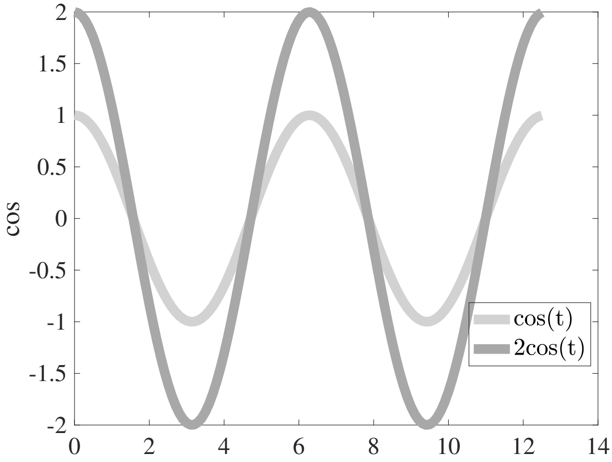
- With $\omega = 1$, the period is $2\pi = 6.28$ and frequency is $\frac{1}{2\pi} = 0.16$.

Figure 15: Cosine waves with ω=1 or 1/2



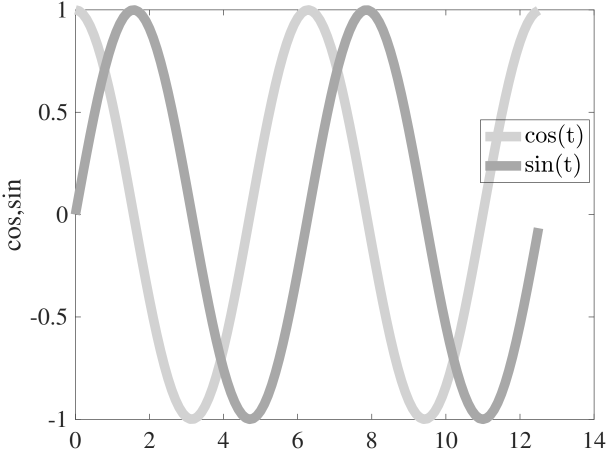
- With $\omega = 1/2$, the period is $4\pi = 12.56$ and frequency is $\frac{1}{4\pi} = 0.08$.

Figure 16: Cosine waves with ω=1 and different amplitudes



- Here are plotted $A\cos(t)$ with $A = 1$ or $A = 2$.

Figure 17: Cosine and Sine waves with ω=1



- $\sin(\omega t)$ behaves the same way, with same amplitude and period, but with a phase shift

- The idea of spectral decomposition is that with sin and cos, we can span the whole space of covariance stationary time series : the typical periodic function is

$$a_\omega \cos(\omega t) + b_\omega \sin(\omega t) \tag{1}$$

whose period is $2\pi/\omega$ but whose phase and amplitude depend on (a_ω, b_ω)
- Here we want to treat a_ω and b_ω as mean zero random variables.
- There is always a sum of type (1) periodic functions that reproduces a given time series
- The spectral density or *spectrum* of a series indicates the weight of each frequency (from low to high) in the total variance of the series.
- These periodic functions are pairwise orthogonal.

- Assume that we observe y_t over T (even) periods, and that it is demeaned.
- Our goal is to decompose y_t into $T/2$ periodic functions of frequencies $\omega_1, \omega_2, ..., \omega_{T/2}$, with
$$\omega_j = \frac{2\pi j}{T}, \quad j = 1, ..., T/2$$
so that period goes from $\frac{2\pi}{\omega_{T/2}} = \frac{2\pi}{\pi} = 2$ to $\frac{2\pi}{\omega_1} = \frac{2\pi}{\frac{2\pi}{T}} = T$
- Then, we want to write y_t as

$$\begin{aligned} y_t &= a_1 \cos(\omega_1 t) + b_1 \sin(\omega_1 t) \\ &+ a_2 \cos(\omega_2 t) + b_2 \sin(\omega_2 t) \\ &+ \cdots \\ &+ a_{T/2} \cos(\omega_{T/2} t) + b_{T/2} \sin(\omega_{T/2} t) \end{aligned} \tag{1}$$

for t=1,...,T.

- Find the T parameters (a_i, b_i) in (2) by OLS.

$$X = \begin{bmatrix} \cos(\omega_1) & sin(\omega_1) & \cdots\cdots & \cos(\omega_{T/2}) & sin(\omega_{T/2}) \\ \cos(2\omega_1) & sin(2\omega_1) & \cdots\cdots & \cos(2\omega_{T/2}) & sin(2\omega_{T/2}) \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \cos(T\omega_1) & sin(T\omega_1) & \cdots\cdots & \cos(T\omega_{T/2}) & sin(T\omega_{T/2}) \end{bmatrix} Y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ \vdots \\ y_{T-1} \\ y_T \end{bmatrix} \beta = \begin{bmatrix} a_1 \\ b_1 \\ \vdots \\ \vdots \\ a_{T/2} \\ b_{T/2} \end{bmatrix}$$

- If we assume $Y = X\beta + u$, we can compute the parameters $\{a_j, b_j\}$.

- The a and b coefficients are computed as

$$\beta = (X'X)^{-1}X'Y$$

- Given that we have T explanatory variables for T observations, the R^2 is one and $u = 0$. Here we are just solving a *representation* problem, not an estimation one.
- Note that the last column of X is a column of 0. It is replaced by a column of 1 to deal with non-centered series.

- Let's consider US hours worked per capita as an example.
- sample 1948Q1-2024Q3, quarterly data
- **306** observations, **153** sine/cosine waves of periodicity 306, 153, 102, 76.5, 61.2, 51, 43.7, 38.2, 34, 30.6, 27.8, 25.5, 23.5, 21.9,. . ., 2 quarters,
- For an in-depth analysis, see Beaudry, Galizia & Portier [2020], "Putting the Cycle Back into Business Cycle Analysis", American Economic Review.

Figure 18: US hours per capita

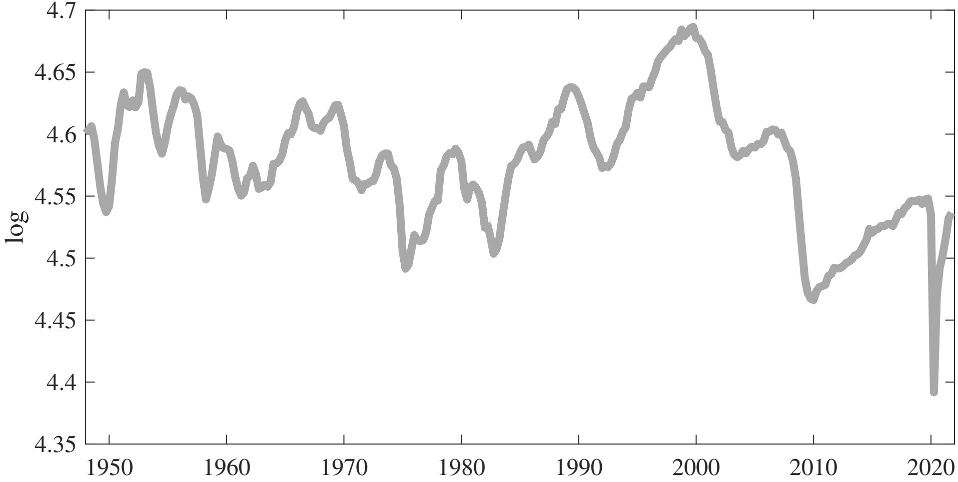


Figure 19: Cosine wave of periodicity 306 quarters

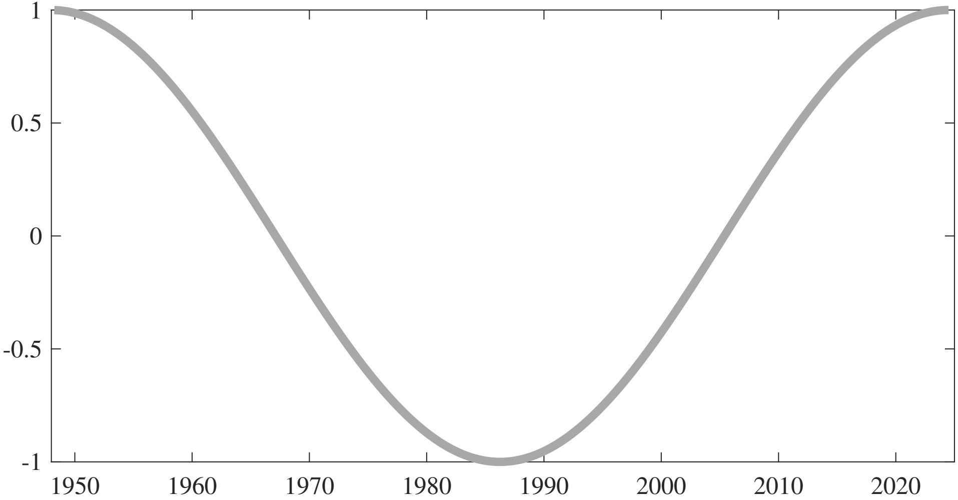


Figure 20: Cosine and sine waves of periodicity 306 quarters

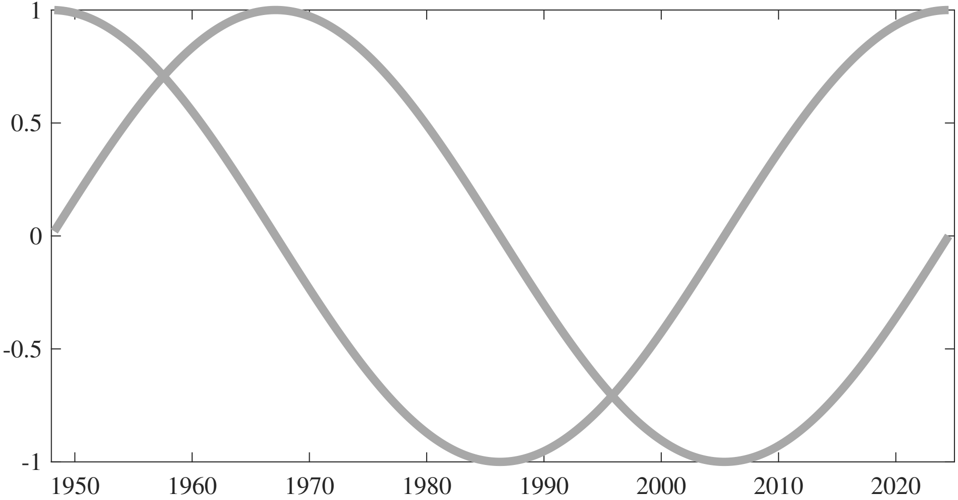


Figure 21: Cosine wave of periodicity 153 quarters

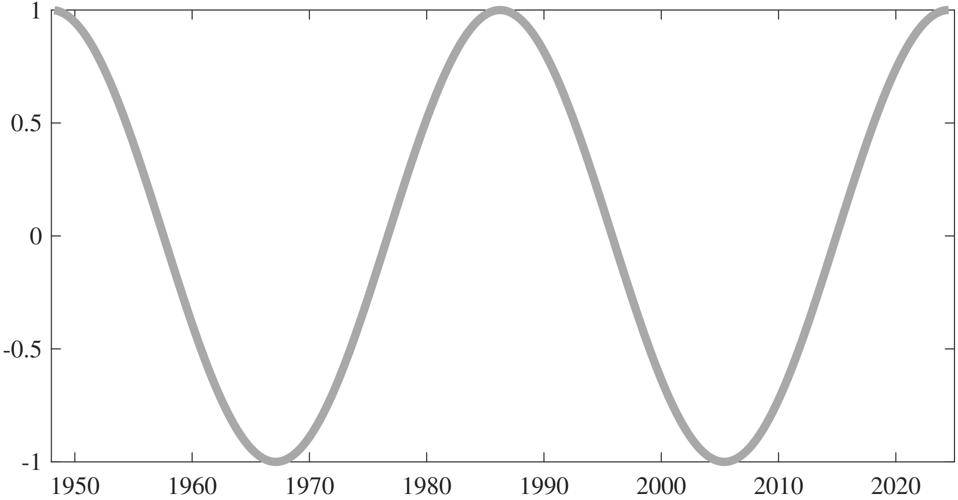


Figure 22: Cosine and sine waves of periodicity 102 quarters

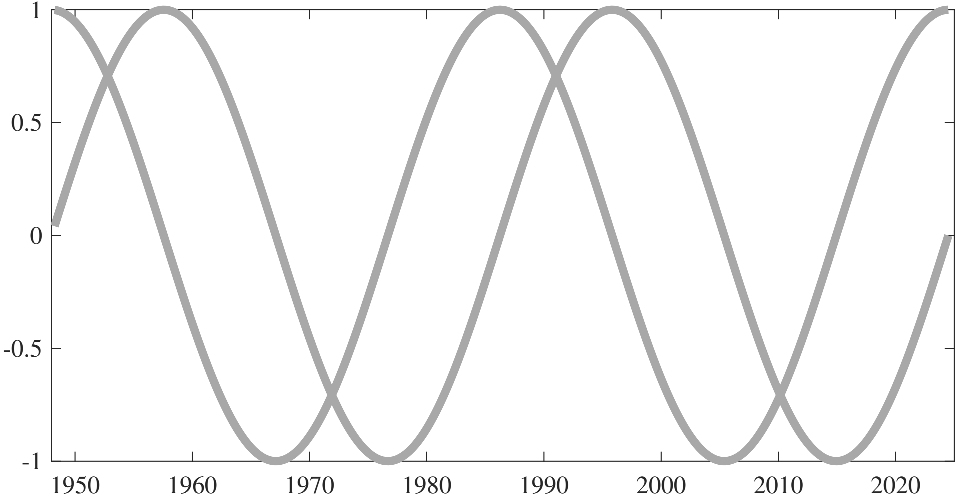


Figure 23: Cosine wave of periodicity 76.5 quarters

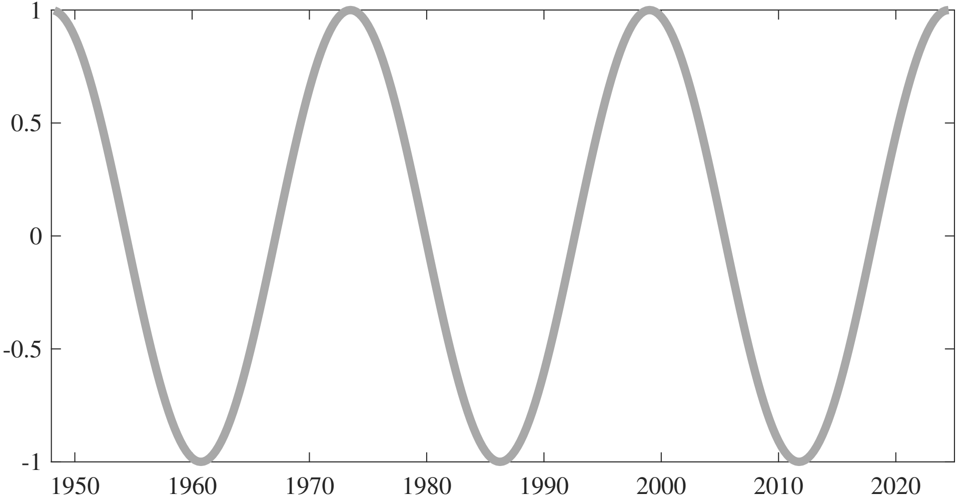


Figure 24: Cosine wave of periodicity 61.2 quarters

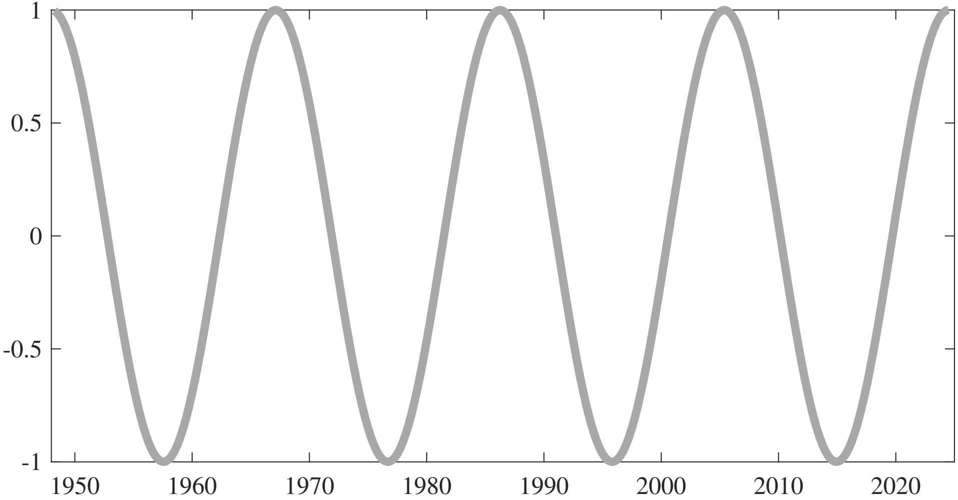


Figure 25: Cosine wave of periodicity 51 quarters

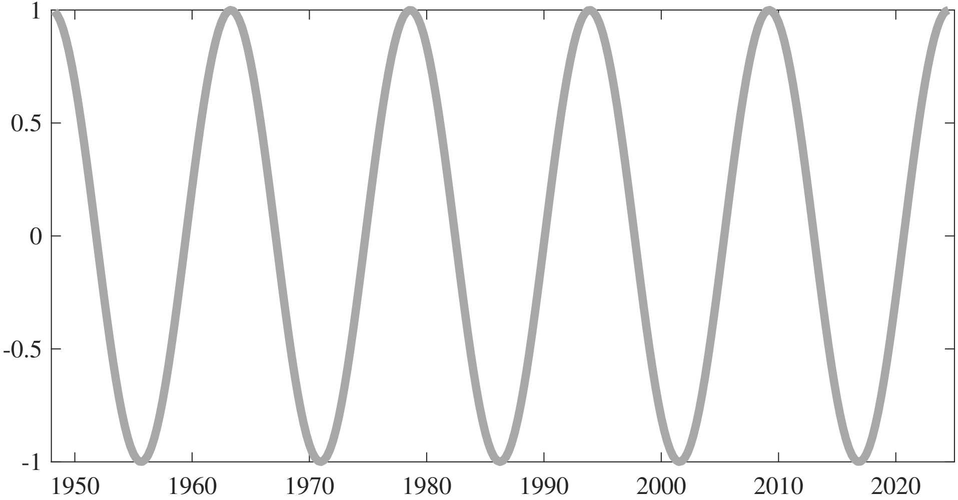


Figure 26: Cosine wave of periodicity 43.7 quarters

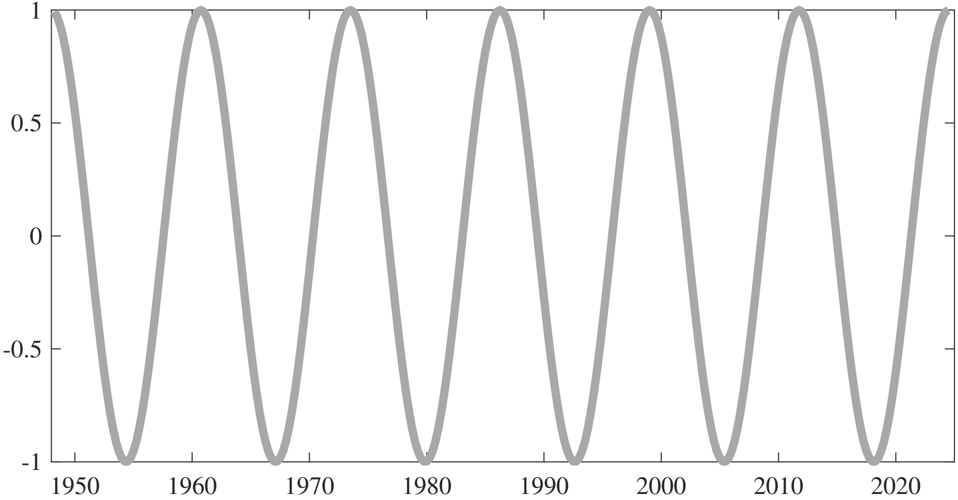


Figure 27: Cosine wave of periodicity 2 quarters

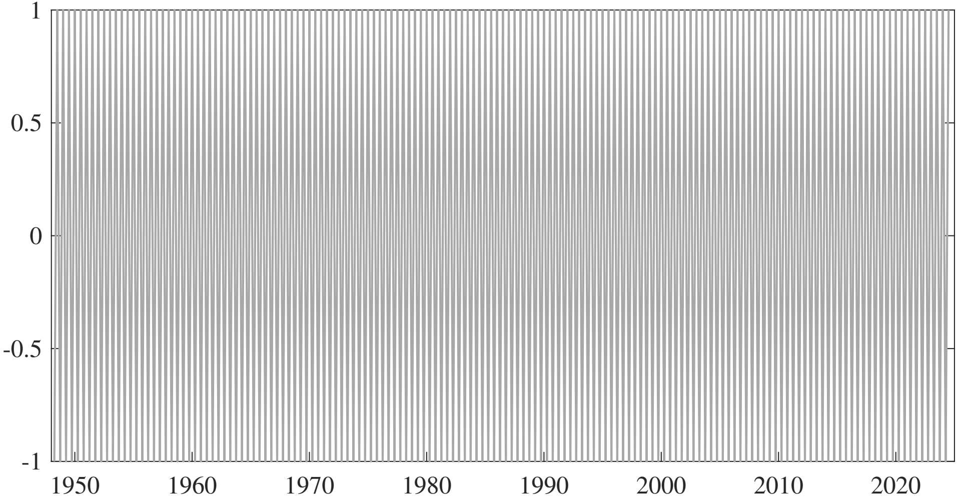


Figure 28: Contribution of Various Periods to Total Variance

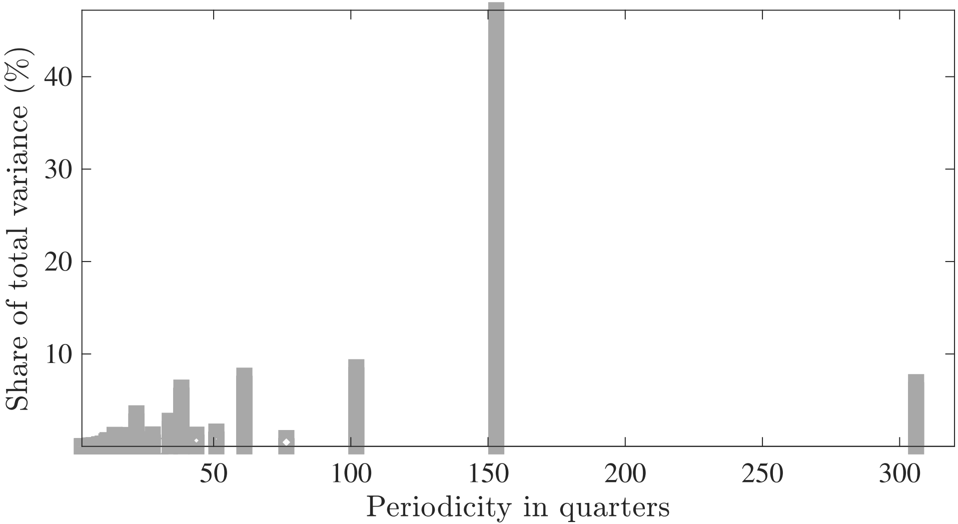


Figure 29: Contribution of Various Periods to Total Variance

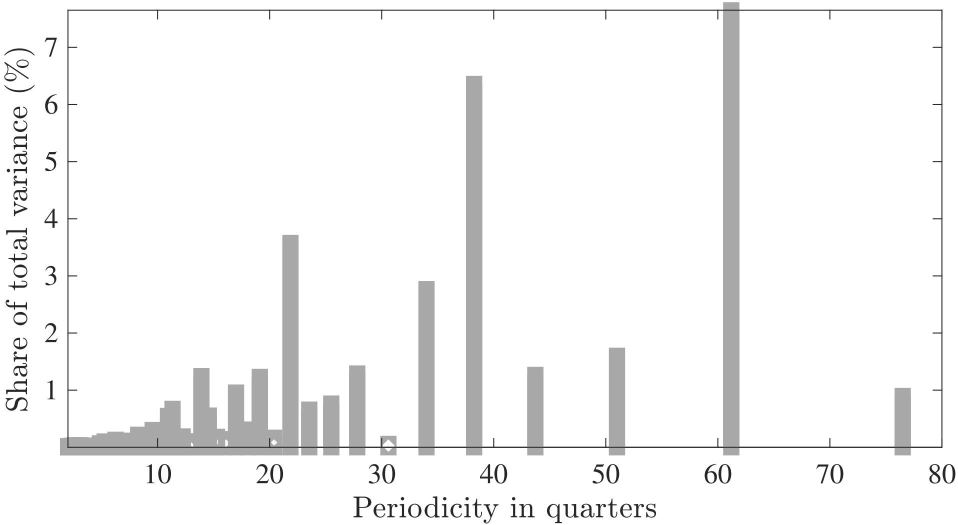


Table 2: Contribution of different frequencies to total variance (ranked from highest to lowest)

Share of Total Variance	Periodicity
47.19	153.0
8.56	102.0
7.65	61.2
6.95	306.0
6.36	38.2
3.58	21.9
2.77	34.0
1.60	51.0
1.29	27.8
1.27	43.7
1.25	13.9
etc	...

Figure 30: Approximation with the First Main Frequency

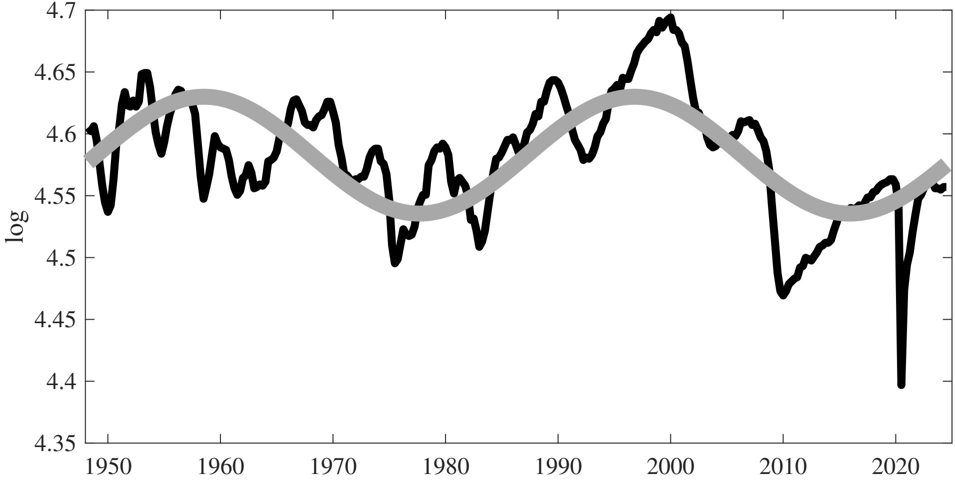


Figure 31: Approximation with the Two Main Frequencies

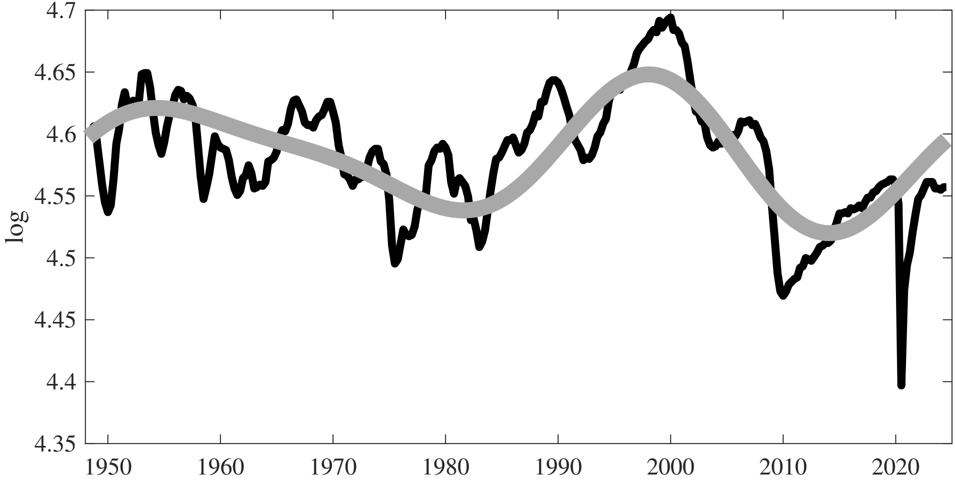


Figure 32: Approximation with the Three Main Frequencies

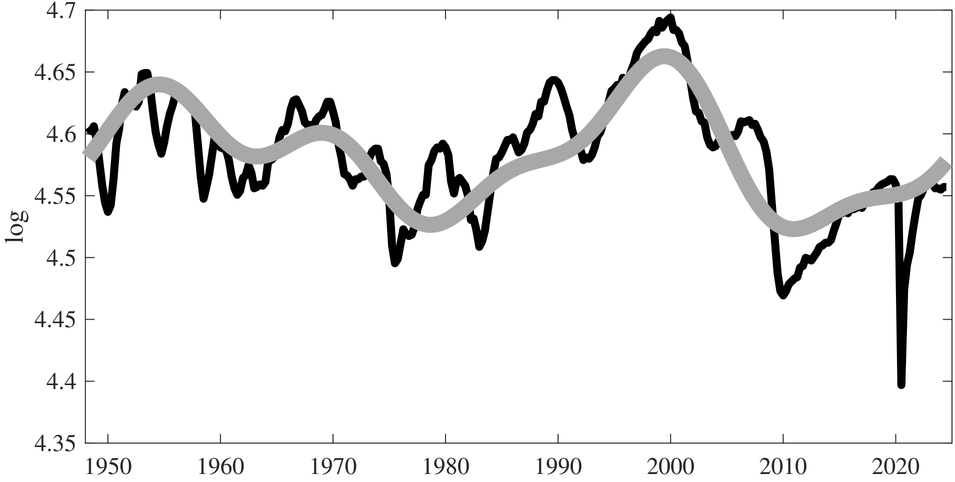


Figure 33: Approximation with the Four Main Frequencies

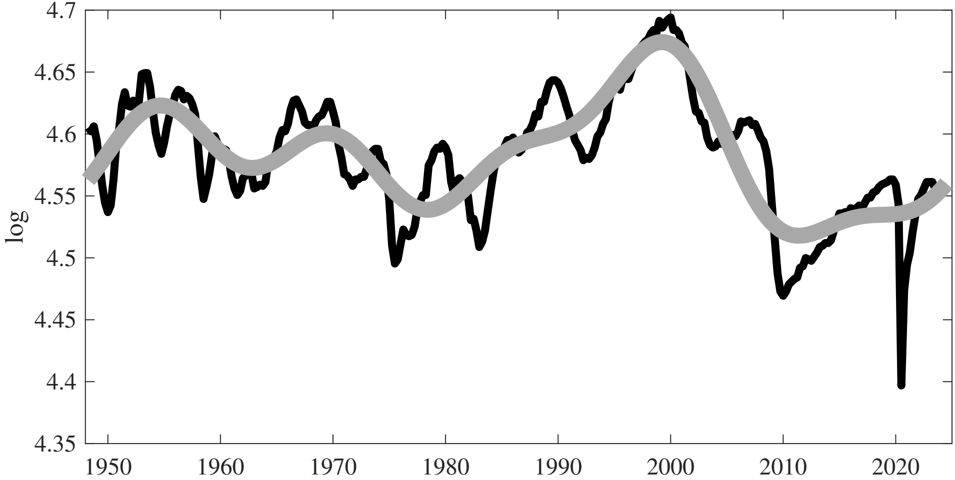


Figure 34: Approximation with the Five Main Frequencies

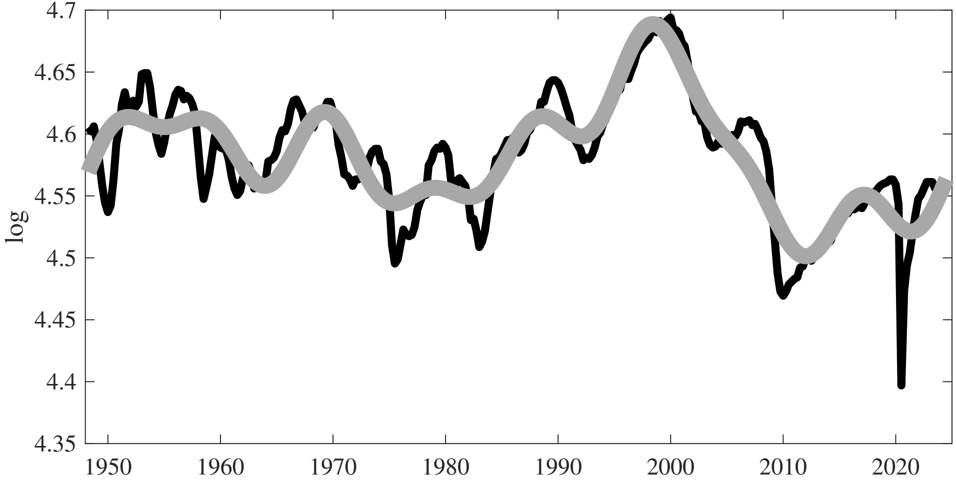


Figure 35: Approximation with the Six Main Frequencies

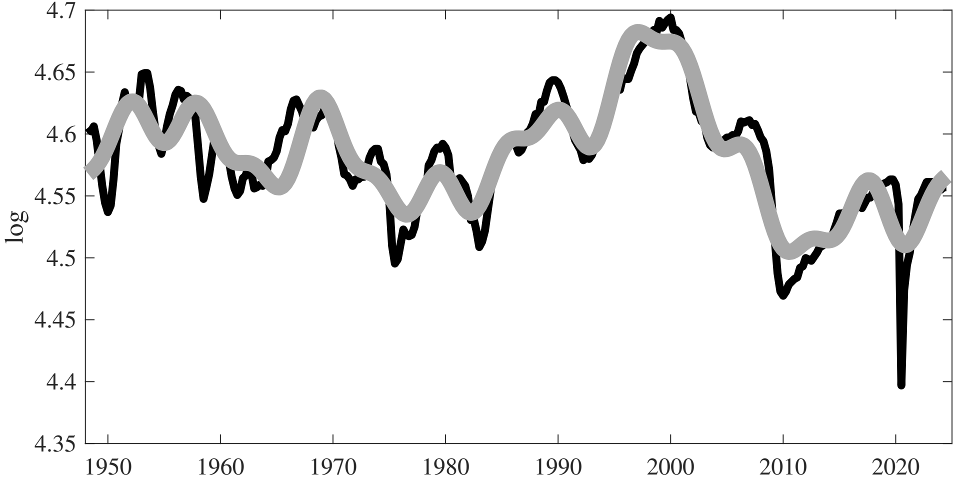


Figure 36: Approximation with the Seven Main Frequencies

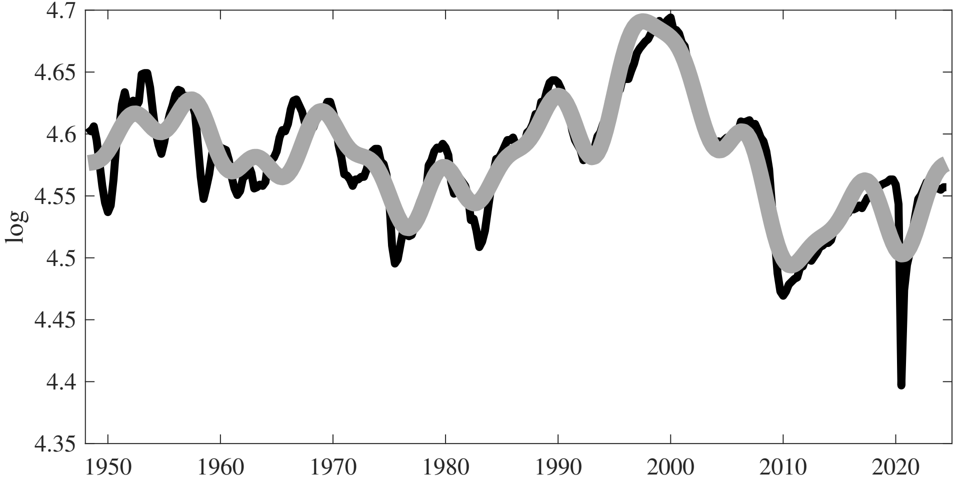


Figure 37: Approximation with the Eight Main Frequencies

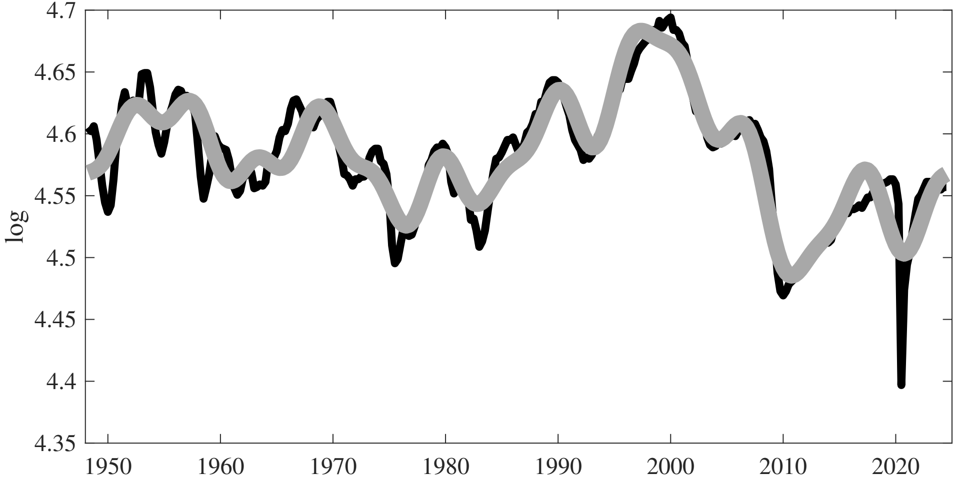


Figure 38: Approximation with the Nine Main Frequencies

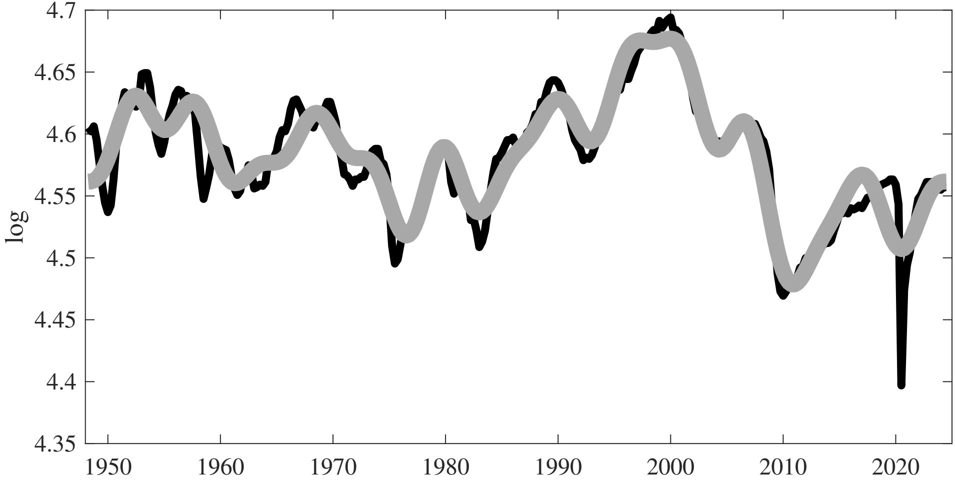


Figure 39: Approximation with the Twenty Main Frequencies

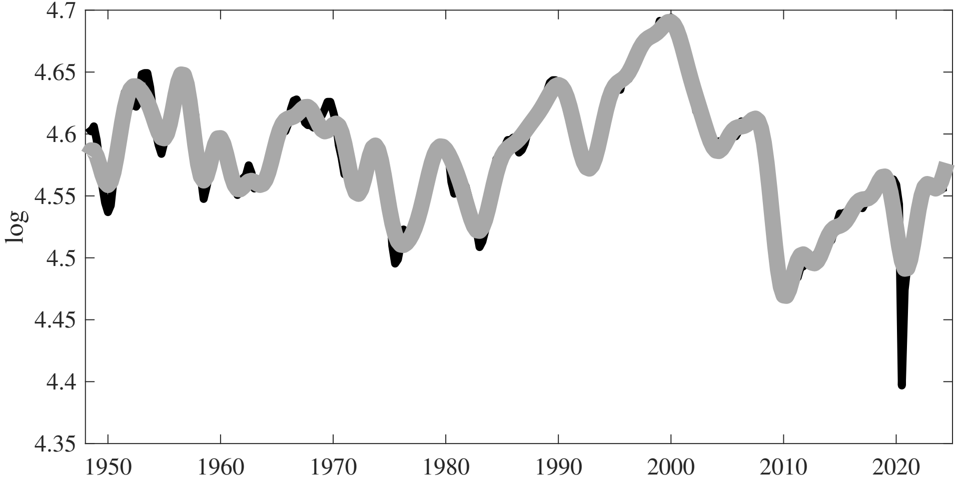


Figure 40: Approximation with the Thirty Main Frequencies

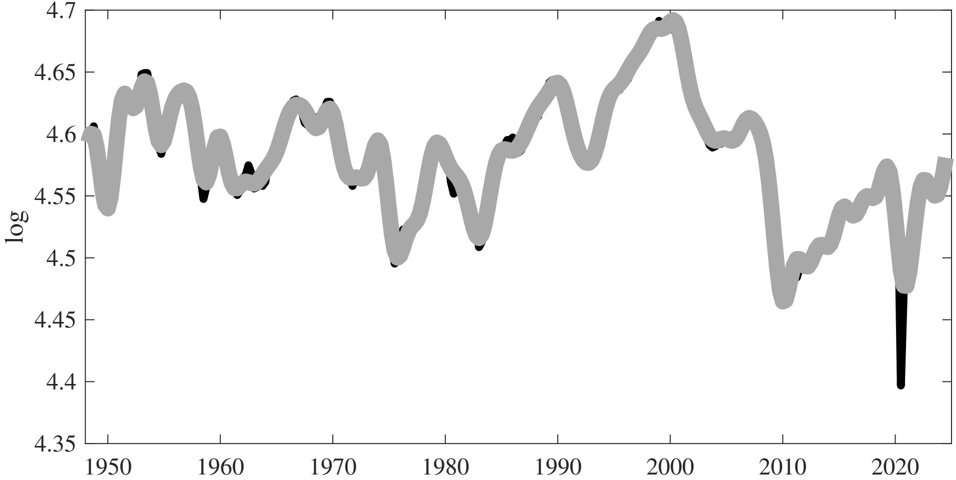
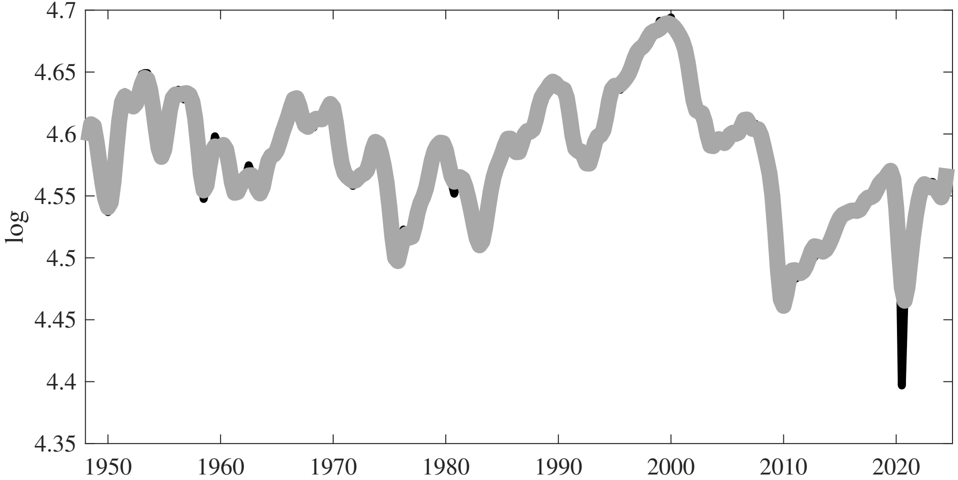


Figure 41: Approximation with the Fourty Main Frequencies



- We can now decide to extract a particular band of frequencies by keeping only the sine and cosine waves at some given periodicities.

$$
\begin{aligned}
y_t &= a_1 \cos(\omega_1 t) + b_1 \sin(\omega_1 t) \\
&+ a_2 \cos(\omega_2 t) + b_2 \sin(\omega_2 t) \\
&+ a_3 \cos(\omega_3 t) + b_3 \sin(\omega_3 t) \\
&+ a_4 \cos(\omega_4 t) + b_4 \sin(\omega_4 t) \\
&+ \cdots \\
&+ a_{T/2} \cos(\omega_{T/2} t) + b_{T/2} \sin(\omega_{T/2} t)
\end{aligned}
$$

- The following figures show
  - × A Band-Pass (6-32) quarters filter, which is quite similar to a Hodrick-Prescott filter with $\lambda = 1600$, and often choen by the literature.
  - × A somewhat different filter (20 to 48 quarters) that removes some low periodicity fluctuations.

Figure 42: Band-Pass filter, 6-32 quarter

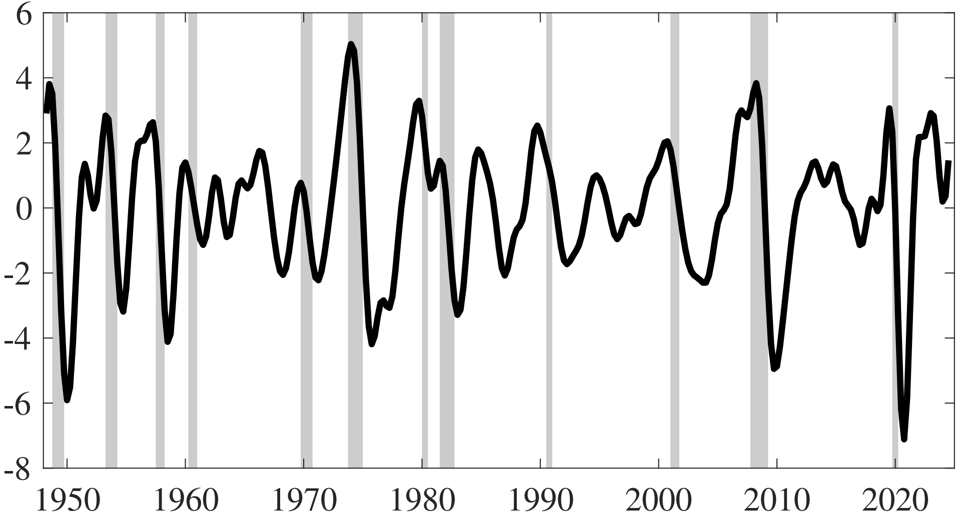


Figure 43: Comparing Band-Pass filter 6-32 quarter and HODRICK-PRESCOTT Filter

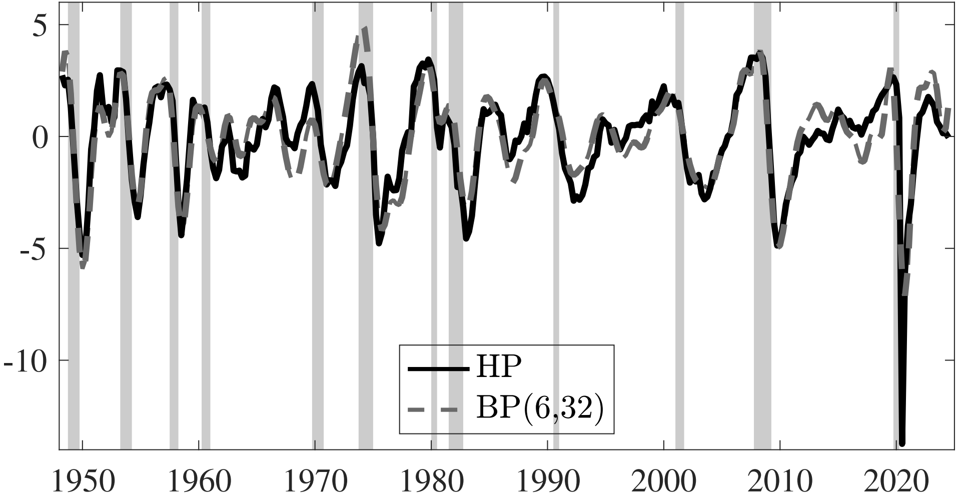


Figure 44: Band-Pass filter, 20-50 quarter

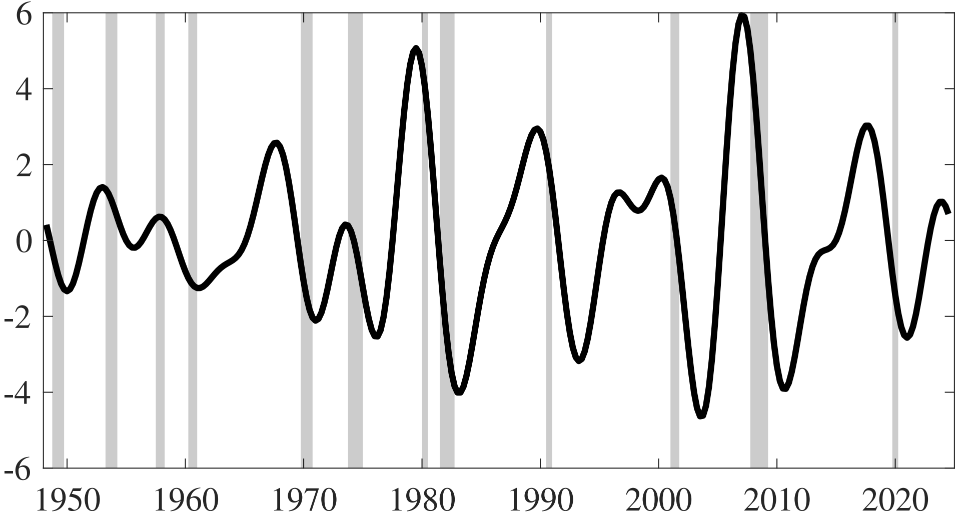
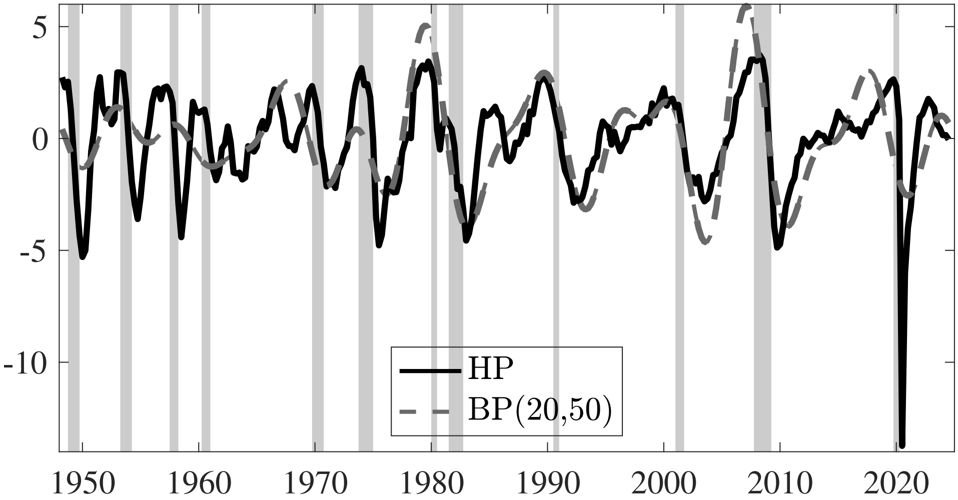


Figure 45: Comparing Band-Pass filter 20-50 quarter and HODRICK-PRESCOTT Filter

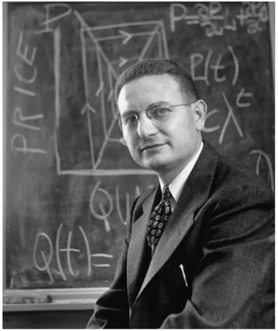


- What do we see?
  - Hodrick-Prescott and Band-Pass (6,32) filters are essentially the same
  - They keep a lot of high frequency-low periodicity fluctuations that are not related with the NBER datation
  - The Band-Pass (20,50) does a better job in fitting the NBER datation and not having higher frequency cycles
  - A model that aims at explaining the NBER cycles should not keep some of the high frequencies generally considered in the literature.

Plan of the Lecture

- No microfoundations
- A dynamic version of the Keynesian cross model

Figure 46: SAMUELSON



- Three equations
- A dynamic version of the Keynesian cross model (constant terms are omitted).
  - National Income Identity: $Y_t = C_t + I_t$
  - Consumption function: $C_t = aY_{t-1} \rightsquigarrow$ multiplier
  - Investment function: $I_t = b(Y_{t-1} - Y_{t-2}) \rightsquigarrow$ accelerator
- which gives a second-order difference equation for $t \geq 2$:

$$Y_t = \underbrace{(a+b)}_{\rho_1} Y_{t-1} + \underbrace{(-b)}_{\rho_2} Y_{t-2}$$

with Y_0 and Y_1 given.

$$Y_t = \underbrace{(a+b)}_{\rho_1} Y_{t-1} + \underbrace{(-b)}_{\rho_2} Y_{t-2}$$

- Steady state is $Y = 0$ (because we have abstracted from constant terms)
- Solution is

  $$Y_t = c_1 \lambda_1^t + c_2 \lambda_2^t$$

  where $\lambda_{1,2}$ are the two roots (real (assumed to be different) or complex) of

  $$\lambda^2 - \rho_1 \lambda - \rho_2 = 0, \qquad (\star)$$

  and $c_{1,2}$ are given by initial conditions: $Y_0 = c_1 + c_2$ and $Y_1 = c_1\lambda_1 + c_2\lambda_2$.
- Which gives $c_1 = \frac{Y_1 - \lambda_2 Y_0}{\lambda_1 - \lambda_2}$ and $c_2 = \frac{\lambda_1 Y_0 - Y_1}{\lambda_1 - \lambda_2}$.

- The model dynamics can be pretty rich, depending on parameters

$$\lambda^2 - \rho_1 \lambda - \rho_2 = 0, \qquad (\star)$$

- If $\rho_1^2 + 4\rho_2 > 0$, the two roots of $(\star)$ are real.
  - × Depending on parameters, the absolute values of λ_i can be
    - both smaller than one $\rightsquigarrow$ convergence to the steady state.
    - one or two larger than one $\rightsquigarrow$ divergence.
- Let's not study the limit case $\rho_1^2 + 4\rho_2 = 0$

$$\lambda^2 - \rho_1\lambda - \rho_2 = 0, \qquad (\star)$$

- If $\rho_1^2 + 4\rho_2 < 0$,
  - the two roots of $(\star)$ are complex and conjugate: $\lambda_{1,2} = \frac{\rho_1 \pm i\sqrt{-(\rho^2+4\rho_2)}}{2} = \alpha \pm \beta i$
  - and they can be rewritten as

$$\lambda_{1,2} = r(\cos\theta \pm i\sin\theta) = re^{\pm}i\theta$$

with $r = \sqrt{\alpha^2 + \beta^2} = (-\rho_2)^{\frac{1}{2}}$ and $\theta = 2\arctan\left(\frac{\beta}{\sqrt{\alpha^2+\beta^2}+\alpha}\right)$

- The solution is

$$y_t = c_1\Big(r\big(\cos(\theta) + i\sin(\theta)\big)\Big)^t + c_2\Big(r\big(\cos(\theta) - i\sin(\theta)\big)\Big)^t$$

- using *de Moivre* theorem:

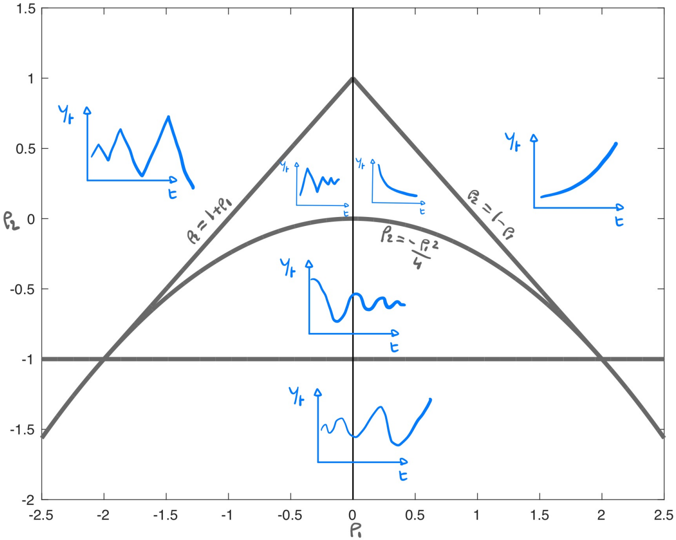
$$\Big(r\big(\cos(\theta) + i\sin(\theta)\big)\Big)^t = r^t\big(\cos(\theta t) + i\sin(\theta t)\big)$$

- we obtain

$$y_t = (-\rho_2)^{\frac{t}{2}}\big(d_1\cos(\theta t) + d_2\sin(\theta t)\big)$$

  with $d_1 = c_1 + c_2$ and $d_2 = (c_1 - c_2)i$
- We have cyclical solution.
- Oscillations can be dampened or explosive depending on $|(-\rho_2)^{1/2}| \lessgtr 1$.

Figure 47: Dynamics as a function of ρ_1 and ρ_2



- Assume

$$Y_t = \rho_1 Y_{t-1} + \rho_2 Y_{t-2} + \varepsilon_t \qquad (\star\star)$$

  and $Y(0) = Y(1) = 0$, $\varepsilon_1 = 1$, $\varepsilon_t = 0$ for $t > 1$
- Let's simulate $(\star\star)$

Figure 48: $\rho_1 = .5$, $\rho_2 = -.9$, $\lambda_1 = .25 + .915i$, $\lambda_2 = .25 - .915i$, $|\lambda_{1,2}| = .9487$

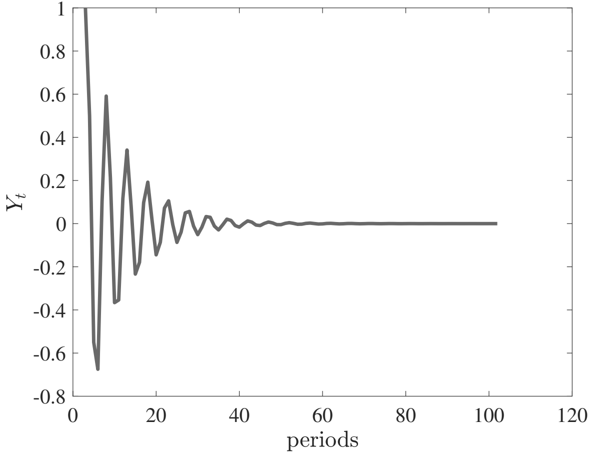


Figure 49: $\rho_1 = .5$, $\rho_2 = -1.2$, $\lambda_1 = .25 + 1.2i$, $\lambda_2 = .25 - 1.2i$, $|\lambda_{1,2}| = 1.095$

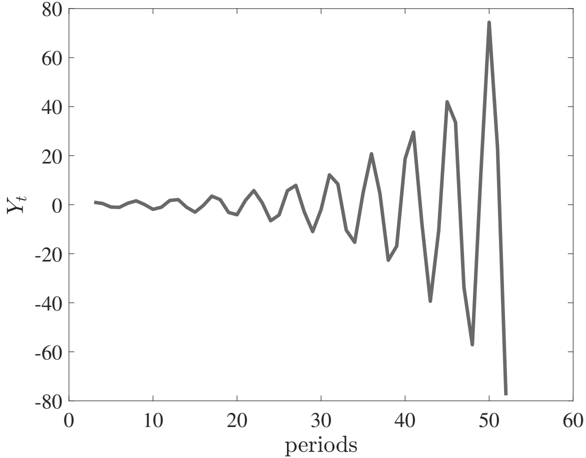


Figure 50: $\rho_1 = .5$, $\rho_2 = -1$, $\lambda_1 = .25 + .9682i$, $\lambda_2 = .25 - 9.682i$, $|\lambda_{1,2}| = 1$

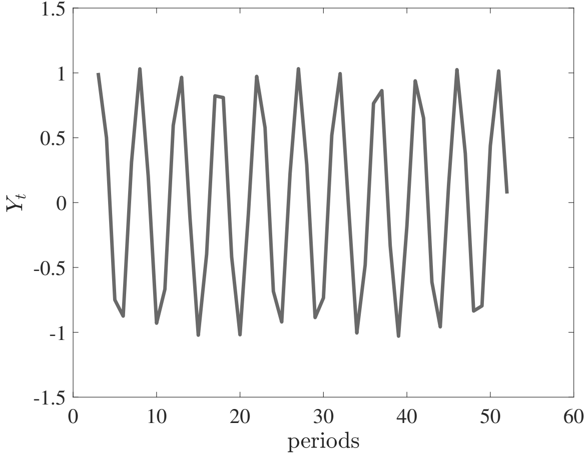


Figure 51: $\rho_1 = .5$, $\rho_2 = .1, \lambda_1 = .65$, $\lambda_2 = -.15$

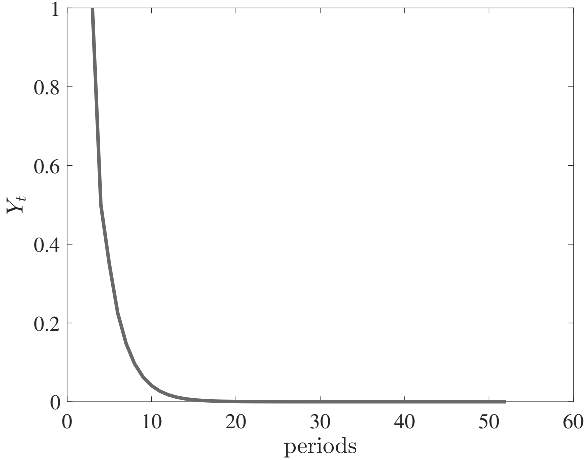


Figure 52: $\rho_1 = 1.1$, $\rho_2 = .1$, $\lambda_1 = 1.18$, $\lambda_2 = -.08$

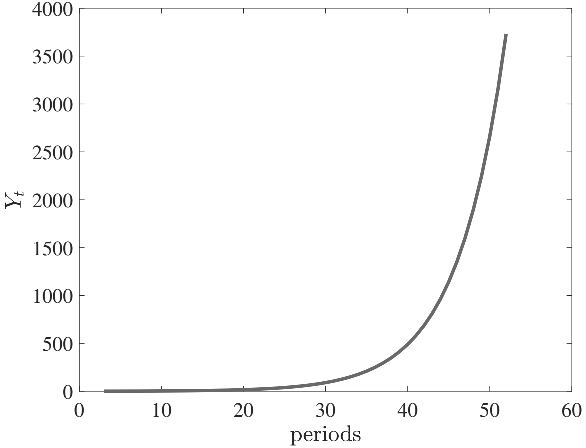
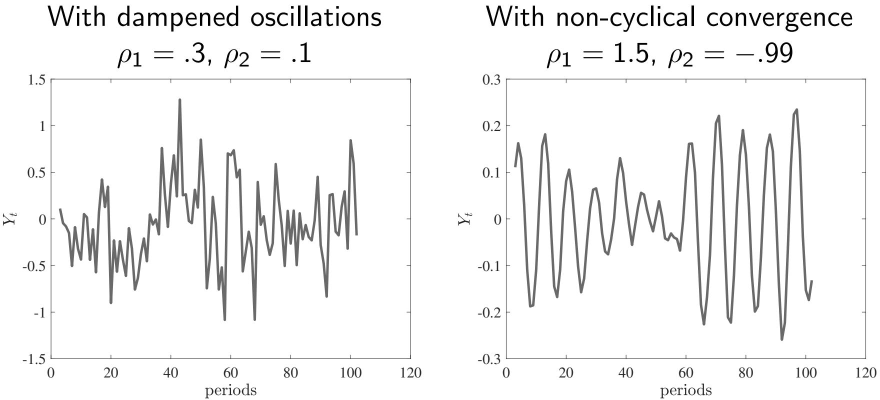


Figure 53: $Y_t = \rho_1 Y_{t-1} + \rho_2 Y_{t-2} + \varepsilon_t$, $\varepsilon_t \rightsquigarrow \mathcal{N}(0, \sigma)$



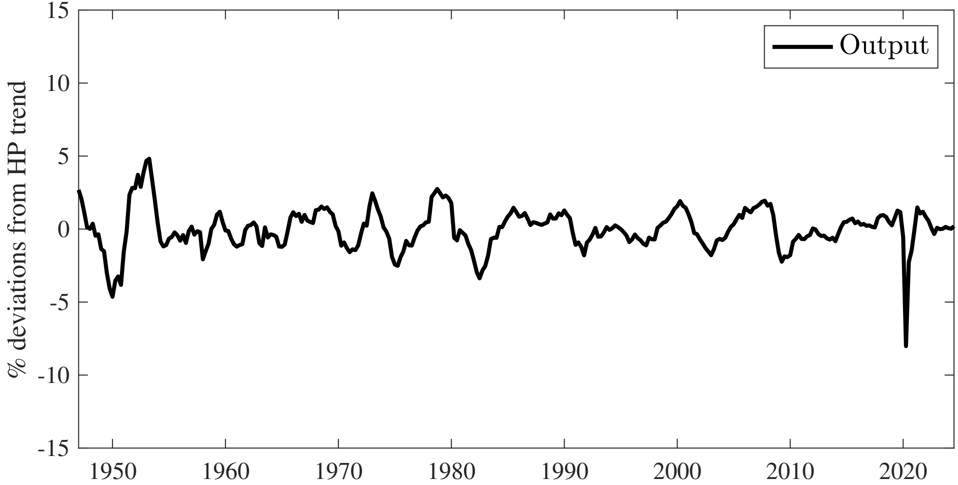
- Illustration of Slutsky [1927] "The summation of random causes as the source of cyclic processes": the moving average of a random series may generate oscillatory movement.

Plan of the Lecture

- We look for **regularities** (*Stylized facts*)
- Business Cycles are characterized by a set of statistics:
  - Volatilities of time series (standard deviations)
  - Comovements of time series (correlations, serial correlations)

- Consumption (C): Nondurables + Services
- Investment (I): Durables + Fixed Investment + Changes in inventories
- Government spending (G)
- Output: $C + I + G$
- Labour: hours worked
- Labour Productivity: Output / labour

Figure 54: Output, Hodrick-Prescott filtered



Main Real Aggregates (1947Q1-2024Q3)

Figure 55: Output and Consumption, HODRICK-PRESCOTT filtered

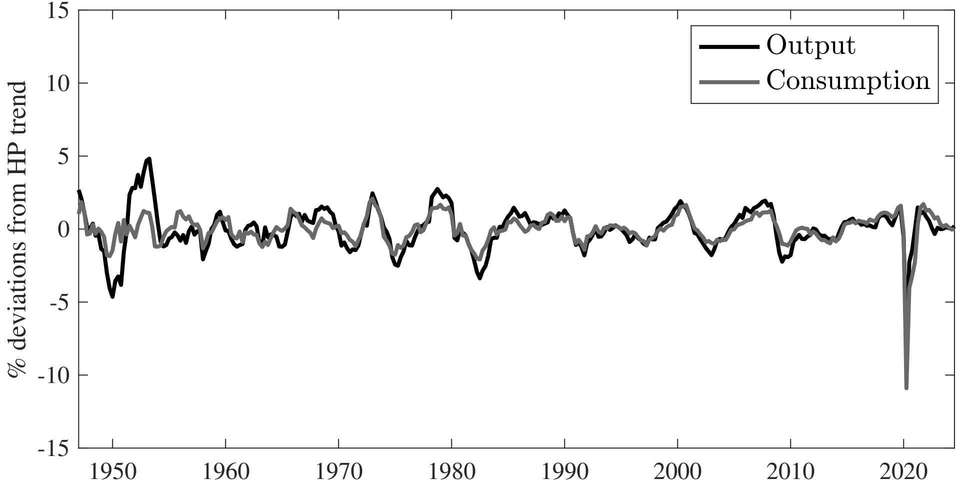


Figure 56: Output, Consumption and Investment, Hodrick-Prescott filtered

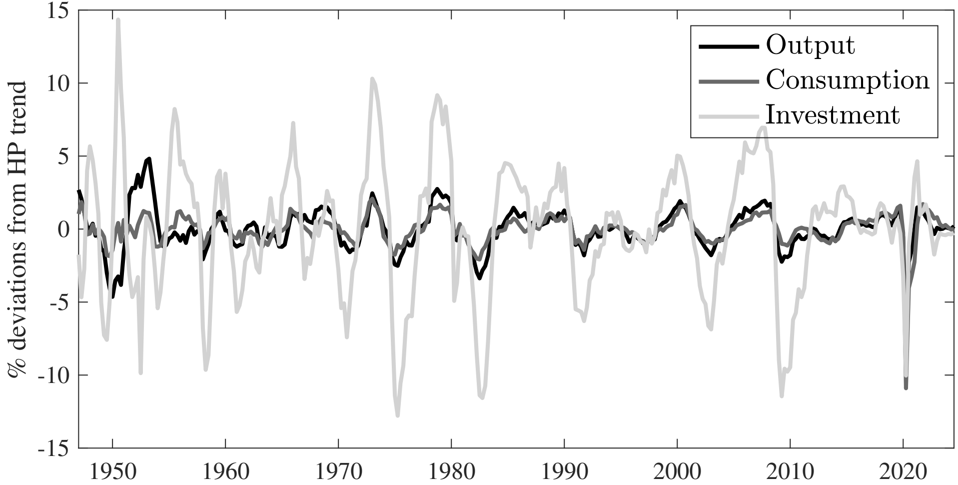


Figure 57: Output and Hours worked, HODRICK-PRESCOTT filtered

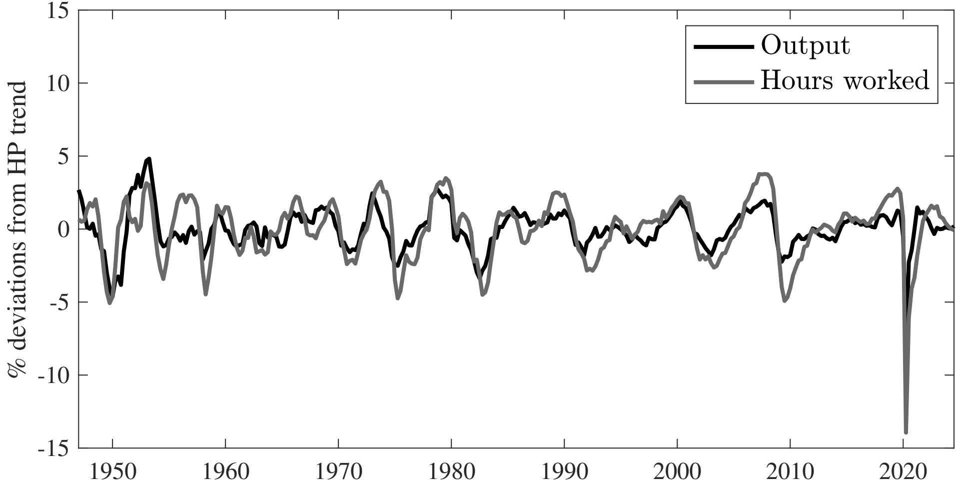
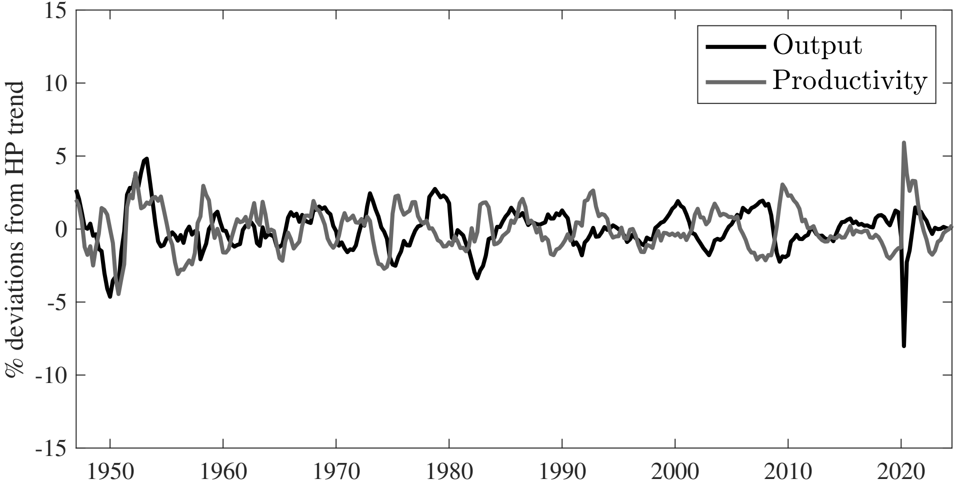


Figure 58: Output and Labour Productivity (Y/H), Hodrick-Prescott filtered



5. Facts

Moments (1947Q1-2024Q3)

- We want to characterize fluctuations $\rightsquigarrow$ amplitude and movements
- Amplitude: volatilities $\rightsquigarrow$ standard deviations
- Comovements: correlations

Table 3: Stylized Facts Using the HODRICK-PRESCOTT Filter

Variable x	$\sigma(x)$	$\sigma(x)/\sigma(y)$	$\rho(x,y)$	$\rho(x,x_{-1})$
Output (y)	1.41	1.00	1.00	0.84
Consumption	1.06	0.75	0.74	0.67
Investment	4.43	3.14	0.51	0.87
Hours worked	2.08	1.47	0.73	0.82
labour productivity	1.42	1.01	-0.08	0.83

- Remark: As far as those moments are concerned, using a Band-Pass(20,50) filter does not affect the results.

Table 4: Stylized Facts Using the Band-Pass (20,48) Filter

Variable x	$\sigma(x)$	$\sigma(x)/\sigma(y)$	$\rho(x,y)$	$\rho(x,x_{-1})$
Output (y)	1.59	1.00	1.00	0.98
Consumption	0.92	0.58	0.76	0.98
Investment	4.42	2.77	0.55	0.98
Hours worked	2.05	1.29	0.75	0.98
labour productivity	1.36	0.86	0.05	0.98

5. Facts

Main Real Aggregates (1947Q1-2024Q3)

Figure 59: Output, Band-Pass (20,48) filtered

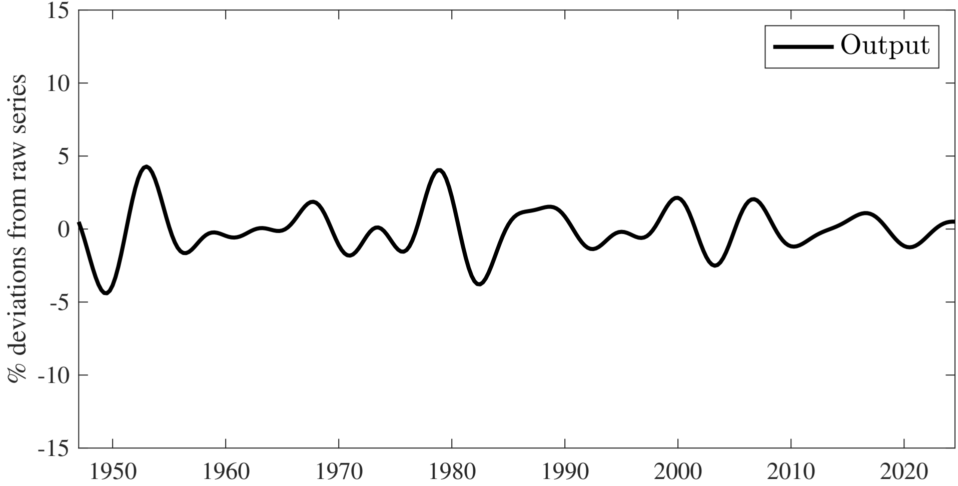


Figure 60: Output and Consumption, Band-Pass (20,48) filtered

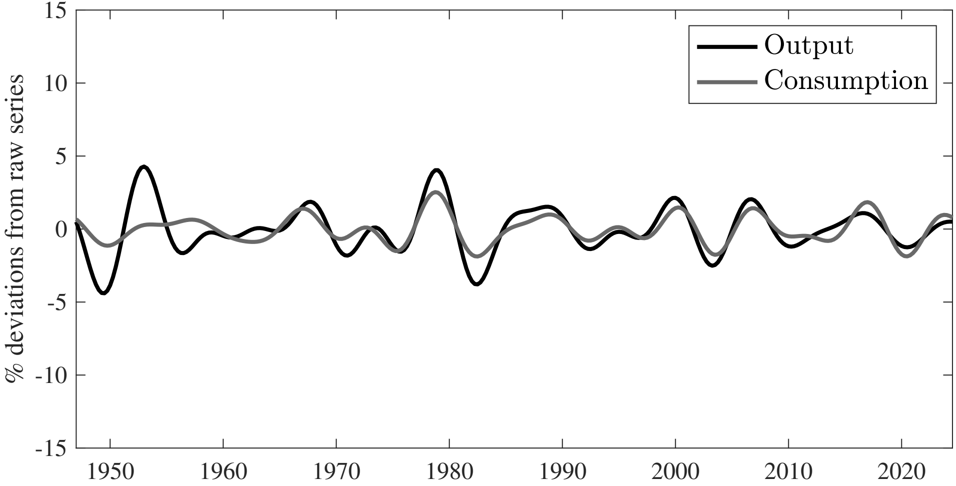


Figure 61: Output, Consumption and Investment, Band-Pass (20,48) filtered

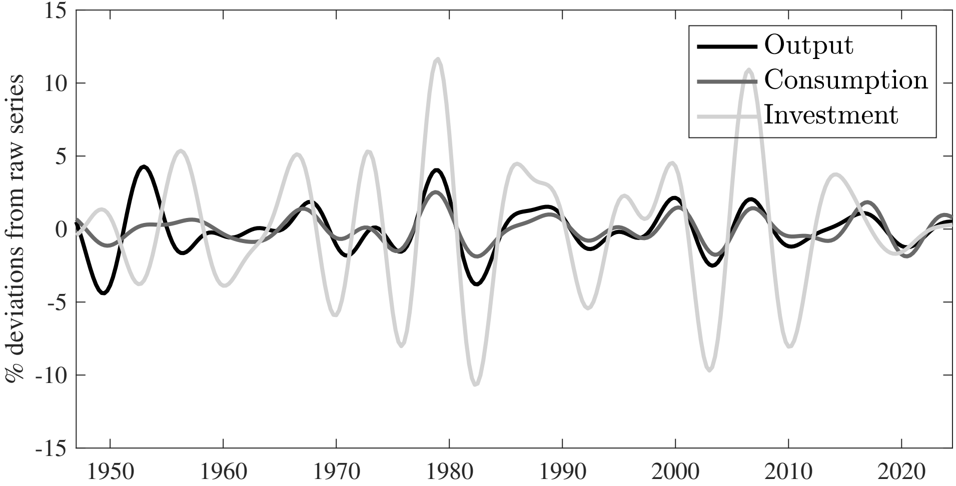
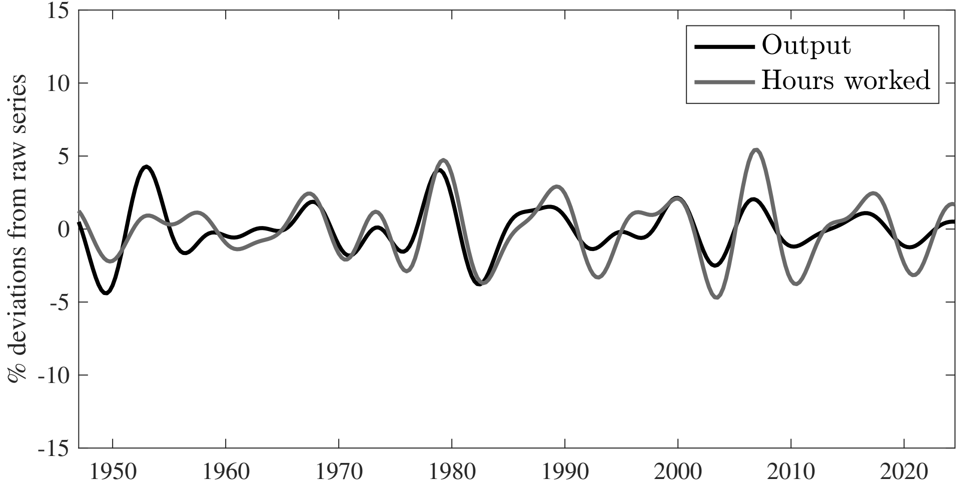
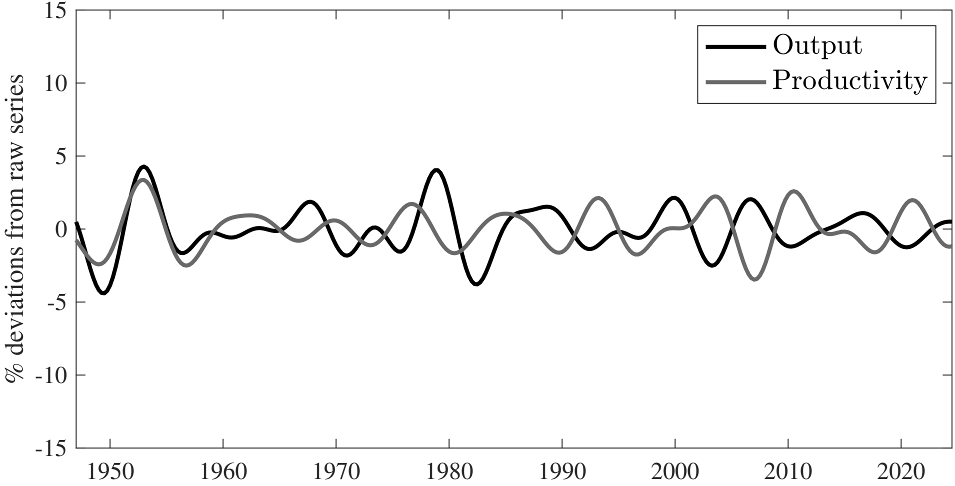


Figure 62: Output and Hours worked, Band-Pass (20,48) filtered



Main Real Aggregates (1947Q1-2024Q3)

Figure 63: Output and Productivity, Band-Pass (20,48) filtered



1. Consumption (of non-durables and services) is less volatile than output
2. Investment is more volatile than output
3. Hours worked are a bit more volatile than output
4. Capital is much less volatile than output (not shown in the table)
5. Labour productivity is about as volatile as output
6. All those variables are persistent and procyclical except labour productivity that is acyclical

Plan of the Lecture

1. Definition
2. Time Domain
3. Frequency Domain
4. A Simple Dynamic Model of the Business cycle
5. Facts about Business Cycles
6. Demand and Supply Shocks in an “UV” model
7. Summary
8. References

- Here I propose a quite general macroeconomic model in General Equilibrium.
- There will be exogenous variables (productivity, economic agents mood. etc...), and I will check if those exogenous variables, when shocked, can create fluctuations that look like a business cycle (meaning that investment, hours, output and consumption are positively correlated)
- N households indexed by j (N very large).
- Two goods: consumption good c (numéraire) and investment good x.
- Two "periods": today, and all the future.
- As such, the model is silent about the dynamics
- A full model would mix the "UV" model with some type of micro-founded SAMUELSON's oscillator.

$$U(c^j, \ell^j) + V(x^j, \theta)$$

- U refers to today utility.
- c is consumption, ℓ is labour.
- $U'_c > 0, U''_c < 0$, $U'_\ell < 0$, $U''_\ell < 0$ and for simplicity we will often assume $U''_{c\ell} = 0$ (U is additively separable in c and ℓ)
- x is investment, which is assumed to be the only form of savings.
- $V(x, \theta)$ is the *continuation value* of savings (investment)
- The V function should be understood as the expected future intertemporal utility of starting the next period with level of savings (investment) x.
- θ represents anything that is affecting the continuation value of investment (changes in expectations, in "optimism", in future productivity of capital, in future taxes, etc...).
- θ is common to all agents
- For simplicity, we will assume $V(x^j, \theta) = \theta V(x^j)$

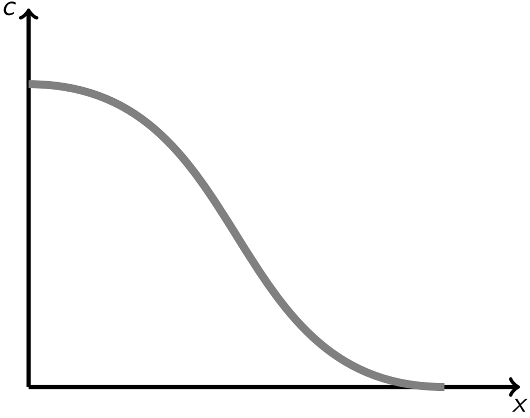
$$H(c, x) = J(\ell; A)$$

- Assume a representative firm
- It produces the consumption and the investment good with labour ℓ,
- A is a technological shifter.
- For a fixed level of input ℓ and technology A, the *production possibility frontier* (ppf) describes the feasible (x, c).
- Slope of the ppf:

$$H_x dx + H_c dc = 0 \Longleftrightarrow \frac{dc}{dx} = -\frac{H_x}{H_c}$$

6. Demand and Supply Shocks in an "UV" model

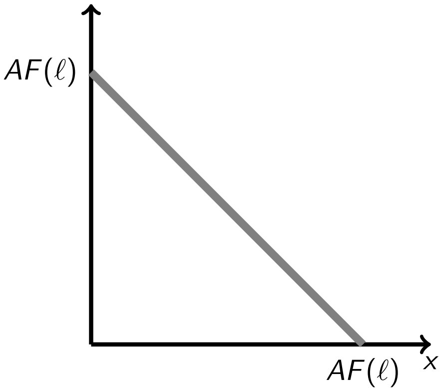
Figure 64: An Arbitrary Production Possibility Frontier



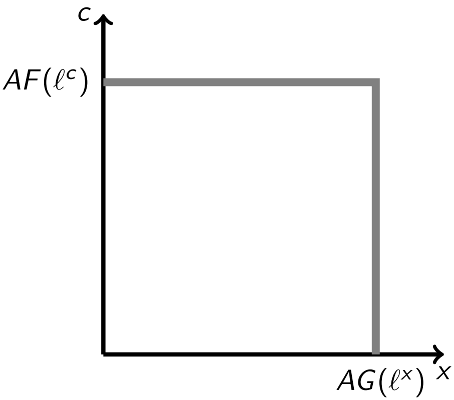
$$H(c, x) = J(\ell; A)$$

- We study use two polar cases:
  - × One good model, one type of labour: $c + x = AF(\ell)$, ("RBC" model)
  - × Two goods model with specialisation of labour: $c = AF(\ell^c)$ and $x = AG(\ell^x)$ ("Gains From Trade"model)

Figure 65: Production Possibility Frontiers



One good model: $c + x = AF(\ell)$



Two goods model with labour specialisation
$c = AF(\ell^c)$ and $x = AG(\ell^x)$

- This RBC model was developed by Kydland & Prescott [1982], "Time to Build and Aggregate Fluctuations".
- Here I present a super simplified version of it
- The initial idea was to model business cycles as the response of the economy to real (productivity) shocks.
- We will study the one good version: $c + x = AF(\ell)$
- There is a representative firm and a representative household ($N = 1$, but competitive behaviour).
- The representative household owns the firm, and receives its profits π.
- Most of what I say is general, but for simplicity I will assume here specific functional forms.
- I assume $AF(\ell) = A\ell^{\alpha}$ with $0 < \alpha < 1$.

- Representative firm: $\max \pi = AF(\ell) - w\ell$ for given w
- FOC:

$$AF'(\ell) = \alpha A \ell^{\alpha - 1} = w, \tag{RBC.1}$$

  where w is the real wage.
- This FOC determines labour demand

- Representative household utility:

$$U(c, \ell) + V(x, \theta) = \gamma \ln c - B\ell + \theta \ln x$$

- Representative household budget constraint:

$$c + x = w\ell + \pi \tag{RBC.0}$$

- Utility maximisation can be written, using the budget constraint to substitute away c:

$$\max_{\ell,x} \gamma \ln \left(w\ell + \pi - x\right) - B\ell + \theta \ln x$$

for w and π given.

$$\max_{\ell,x} \gamma \ln \underbrace{(w\ell + \pi - x)}_{c} - B\ell + \theta \ln x$$

- FOC wrt ℓ: $\frac{\gamma w}{c} = B$.
- FOC wrt x: $\frac{\gamma}{c} = \frac{\theta}{x}$.

- General equilibrium is given by quantities c, ℓ, x, wage w and profit π such that
  1. for given w and π, quantities maximize utility and profits, and therefore satisfy the FOCs
  $$\alpha A\ell^{\alpha-1} = w, \tag{RBC.1}$$
  $$\frac{\gamma w}{c} = B, \tag{RBC.2}$$
  $$\frac{\gamma}{c} = \frac{\theta}{x}, \tag{RBC.3}$$
  and the budget constraint (RBC.0);
  2. labour market clears (ℓ is the same for the representative firm and household), good market clears
  $$c + x = A\ell^{\alpha}, \tag{RBC.4}$$
  and profits are given by
  $$\pi = A\ell^{\alpha} - w\ell. \tag{RBC.5}$$
- Note that if (RBC.4) and (RBC.5) hold, then the household budget constraint is satisfied (this is a version of the WALRAS's law)
- Exogenous variables that we will shock are A, θ, B, γ.

- Now let's compute equilibrium quantities c, x, ℓ
- (RBC.3) gives

$$c = \frac{\gamma}{\theta} x, \tag{RBC.6}$$

- (RBC.1) and (RBC.2) give $\ell = \left(\frac{B}{\alpha\gamma A} c\right)^{\frac{1}{\alpha-1}}$. Substitute c by its expression in (RBC.6) to get

$$\ell = \left(\frac{B}{\alpha A \theta} x\right)^{\frac{1}{\alpha-1}} \tag{RBC.7}$$

- Now replace c and ℓ in (RBC.4) using (RBC.5) and (RBC.6) to obtain an equation in which x is the only unknown:

$$\underbrace{\frac{\gamma}{\theta} x}_{c} + x = A \left(\underbrace{\left(\frac{B}{\alpha A \theta} x\right)^{\frac{1}{\alpha-1}}}_{\ell} \right)^{\alpha}$$

- This last equation can be solve for x (divide both sides by x) and the solution is

$$\boxed{x = A\left(\frac{\theta}{(\gamma+\theta)^{1-\alpha}}\right)\left(\frac{\alpha}{B}\right)^{\alpha}}$$

- We can use the above equation in (RBC.5) and (RBC.6) to obtain the equilibrium values of c and ℓ.

$$\boxed{c = A\left(\frac{\gamma}{(\gamma+\theta)^{1-\alpha}}\right)\left(\frac{\alpha}{B}\right)^{\alpha}}$$

$$\boxed{\ell = \frac{\alpha(\gamma+\theta)}{B}}$$

- Finally, using $y = A\ell^{\alpha}$, we obtain

$$\boxed{y = A\left(\frac{\alpha(\gamma+\theta)}{B}\right)^{\alpha}}$$

- Using the above solution , one can compute the sign of the response of x, c, ℓ, y to shocks A, B, θ, γ, and see if those shock can create a business cycle (x, c, ℓ, y moving in the same direction)
- The signs are given in the table below.

	∂c	$\partial \ell$	∂x	∂y
$\partial A > 0$	$+$	0	$+$	$+$
$\partial \theta > 0$	-	$+$	$+$	$+$
$\partial B > 0$	-	-	-	-
$\partial \gamma > 0$	$+$	$+$	-	$+$

- Let's discuss those results

A “Real Business Cycle” Model

$c + x = AF(\ell)$

	∂c	$\partial \ell$	∂x	∂y
$\partial A > 0$	$+$	0	$+$	$+$

- $\partial A > 0$ represents a positive technology shock:
  - The technological shock increases labour productivity, which increases labour supply (substitution effect) and also increase agents wealth (wealth effect), which increases consumption and leisure, and therefore decrease labour supply.
  - With these functional forms, the wealth and substitution effects exactly offset each other, so that ℓ stays constant. It is possible to obtain $\partial \ell > 0$ by changing the preferences (at the cost of loosing the analytical solution).
  - c and x increase, so that y also increases.
  - Therefore, productivity shocks can create business cycle movements ... but the data show that productivity is not positively correlated with the cycle ... productivity shocks cannot be the main driver of business cycles in that model.
  - On top of that, it is not clear that we do observe ongoing technology shocks at business cycle frequency.

	∂c	$\partial \ell$	∂x	∂y
$\partial\theta > 0$	-	$+$	$+$	$+$

- $\partial\theta > 0$ represents a positive shock to the future value of investment. It captures all the possible changes in expectations, mood of entrepreneurs, news about future productivity, etc...
  - × Everything else equal, economic agents value more investment, and therefore increase their demand for investment ($\partial x > 0$)
  - × This increase in x is financed by working more (so that $\partial \ell > 0$ and $\partial y > 0$) ...
  - × .. but also by consuming less $\partial c < 0$.
  - × c and x therefore move in opposite direction: demand shocks cannot be the main driver of business cycles in that model.

6. Demand and Supply Shocks in an "UV" model

A "Real Business Cycle" Model

$U(c, \ell) + V(x, \theta) = \gamma \ln c - B\ell + \theta \ln x$

	∂c	$\partial \ell$	∂x	∂y
$\partial B > 0$	-	-	-	-

- $\partial B > 0$ represents a "laziness" shock:
  - × Everything else equal, economic agents value more leisure, and therefore decrease hours worked, consumption and investment.
  - × This is creating a business cycle-like movements: all variables comove...
  - × .. but it is hard to think of laziness shocks as a relevant story for macroeconomic fluctuations.
  - × In a richer model with taxes on labour income, an increase in taxes acts as an increase in B.

	∂c	$\partial \ell$	∂x	∂y
$\partial\gamma>0$	$+$	$+$	-	$+$

- $\partial\gamma>0$ represents a positive demand (for consumption) shock :
  - Everything else equal, economic agents value more consumption, and therefore decide to consume more.
  - They finance that increase in c by working more and investing less.
  - Therefore, consumption and investment move in opposite direction...
  - .. so that demand shocks of that type cannot produce business cycle fluctuations in that model

- As we have seen, the "Real Business Cycle" cannot propose a relevant theory of business cycles because
  - × The shocks that create comovements are either unrealistic (laziness shocks) or not correlated with the cycle (technology shocks)
  - × The shocks that are more realistic (consumption and investment demand shocks) always move investment and consumption in opposite direction, which is not what we see in the business cycle.
  - × We need a different model.
  - × One possible model is the RBC model augmented with sticky prices (see Problem Set 2), in the line of what we will study later in the "Liquidity Trap" lecture.
  - × Here I want to explore another model that explicitly model trade linkages between agents.

- We keep working in the UV framework.
- But we now consider two sectors (consumption and investment) and two agents that are producing only in one of the two sectors.
- Because they both like consumption and investment, they will be gains from trade between each other. I will call this model a "Gains From Trade" (GFT) model.
- Shocks, by affecting the terms of trade between the two agents, will generate business cycles.
- Here again, most of the results are pretty general, but it is easier to derive them with specific functional forms
- This model is taken from Beaudry & Portier [2014], "Understanding Noninflationary Demand-Driven Business Cycles" , NBER Macroeconomics Annual, University of Chicago Press.

- There are two firms, each one being the representative firm of its sector:
  - $c = A\ell^c$ in the consumption sector (the numéraire good)
  - $x = A\ell^x$ in the investment sector (price q)
- Wages are w^c and w^x.
- Profits are $\pi^c = A\ell^c - w^c\ell^c$ and $\pi^x = qA\ell^x - w^x\ell^x$
- Because of perfect competition and constant returns, firms will make zero profit, and we will have in equilibrium

$$w^c = A \tag{GFT.1}$$

$$w^x = q\,A \tag{GFT.2}$$

- There are two households, a representative consumption-worker and a representative investment-worker.
- Each household works only in one of the two sectors.
- They both have UV preferences.
- For tractability, we assume the following preferences
  - c-worker: $\ln c^c - B^c \ell^c + \theta \ln x^c$
  - x-worker: $\ln(c^x - B^x (\ell^x)^2) + \theta \ln x^x$
- Their budget constraints are (I am omitting profits because they are lump-sum payments and zero in equilibrium)
  - c-worker:
  $$c^c + qx^c = w^c \ell^c \tag{GFT.3}$$
  - x-worker:
  $$c^x + qx^x = w^x \ell^x \tag{GFT.4}$$

- Substituting consumption away in the utility using the budget constraint, the utility maximization problem of the two households can be written as:
  - c-worker:
$$\max_{\ell^c, x^c} \ \ln(w^c \ell^c - q x^c) - B^c \ell^c + \theta \ln x^c$$
  - x-worker:
$$\max_{\ell^x, x^x} \ \ln\left(w^x \ell^x - q x^x - B^x (\ell^x)^2\right) + \theta \ln x^x$$

- First Order Conditions are:
  - c-worker:
    - wrt ℓ^c:

$$\frac{w^c}{c^c} = B^c \tag{GFT.5}$$

    - wrt x^c:

$$\frac{q}{c^c} = \frac{\theta}{x^c} \tag{GFT.6}$$

  - x-worker:
    - wrt ℓ^x:

$$\frac{w^x - 2B^x\ell^x}{c^x - B^x(\ell^x)^2} = 0 \tag{GFT.7}$$

    - wrt x^x:

$$\frac{q}{c^x - B^x(\ell^x)^2} = \frac{\theta}{x^x} \tag{GFT.8}$$

- General equilibrium of the GFT model is given by quantities $c, x, \ell, c^c, x^c, \ell^c$, c^x, x^x, ℓ^x and prices w^c, w^x, q such that
  1. given prices the quantities maximize utility and profits, and therefore satisfy the FOCs (GFT.5) to (GFT.8), together with budget constraints (GFT.3) and (GFT.4)
  2. prices are such that markets clear, namely labour markets ((GFT.1) and (GFT.2)) (ℓ^c and ℓ^x are the same for firms and households), consumption and investment market clear:

$$c^c + c^x = c = A\ell^c, \tag{GFT.9}$$

$$x^c + x^x = x = A\ell^x. \tag{GFT.10}$$

- Exogenous variables that we will shock are A, θ.

- Now we need to solve the model for the equilibrium.
- First, note that in equilibrium, $w^c \ell^c = A\ell^c = c$.
- Using the budget constraint of the c-worker (GFT.3) and the definition of c in (GFT.9), we obtain $c^c + q\,x^c = c = c^c + c^x$, which gives

$$q\,x^c = c^x \tag{GFT.11}$$

- This is a key equation that shows the amount of trade between the $c-$ and $x-$ workers.
- The $x-$worker is producing x, but wants to invest and consume. She is therefore trading a part of her production $q\,x^c$ against some consumption goods c^x. This is agreed by the $c-$worker that would like not only to consume, but also to invest.
- Equation (GFT.11) show that trade is balanced between the two workers

6. Demand and Supply Shocks in an "UV" model

A "Gains From Trade" Model

- using (GFT.1) and (GFT.2), we replace everywhere w^c with A and w^x with $q\,A$
- (GFT.5) writes $\boxed{c^c = \frac{A}{B^c}}$, which is the equilibrium value for c^c
- (GFT.6) rewrites $q\,x^c = \frac{\theta A}{B^c}$, so that (GFT.11) gives the equilibrium value of c^x: $\boxed{c^x = \frac{\theta A}{B^c}}$.
- Using the equilibrium values of c^x and c^c, (GFT.9) gives $\boxed{\ell^c = \frac{(1+\theta}{B^c}}$
- As $c = A\ell^c$, we have $\boxed{c = \frac{A(1+\theta}{B^c}}$
- (GFT.8) can be rewritten

$$q\,x^x = \theta c^x - \theta B^x(\ell^x)^2 \qquad (GFT.12)$$

(GFT.4) implies $q\,x^x = q\,A\ell^x - c^x$, which we can put in (GFT.12) to obtain

$$qA\ell^x + \theta B^x(\ell^x)^2 = (1+\theta)c^x \qquad (GFT.13)$$

- (GFT.7) implies $qA = 2B^x\ell^x$. Substituting in (GFT.13) and using the equilibrium value for c^x, we obtain the equilibrium level of ℓ^x: $\boxed{\ell^x = \sqrt{\frac{(1+\theta)\theta A}{(2+\theta)B^x B^c}}}$
- Then given $x = A\ell^x$, we obtain equilibrium total investment $\boxed{x = \sqrt{\frac{(1+\theta)\theta A^3}{(2+\theta)B^x B^c}}}$.
- Total hours worked are $\ell = \ell^c + \ell^x$, so that $\boxed{\ell = \frac{1+\theta}{B^c} + \sqrt{\frac{(1+\theta)\theta A}{(2+\theta)B^x B^c}}}$.
- The relative price of investment is obtained from (GFT.7): $q = \frac{2B^x\ell^x}{A}$, so that, using equilibrium value for ℓ^x, we obtain $\boxed{q = \sqrt{\frac{4(1+\theta)\theta B^x}{(2+\theta)AB^c}}}$.
- Finally, GDP is $y = c + q\,x$, which gives $\boxed{y = \frac{A(1+\theta)}{B^c} + \frac{2(1+\theta)\theta A}{(2+\theta)B^c}}$.

- From the solutions (the framed equations), we can compute the impact on equilibrium variables of a technological shock $\partial A > 0$ and from a demand shock $\partial\theta > 0$.
- Those impacts are summarized in the table below.

	∂c	∂x	$\partial \ell^c$	$\partial \ell^x$	$\partial \ell$	∂q	∂y
$\partial A > 0$	$+$	$+$	0	$+$	$+$	-	$+$
$\partial\theta > 0$	$+$	$+$	$+$	$+$	$+$	$+$	$+$

- Main property: In the GFT model, both demand and supply shock create business cycle-like comovements.

6. Demand and Supply Shocks in an "UV" model

A "Gains From Trade" Model

$c = A\ell^c$, $x = A\ell^x$

	∂c	∂x	$\partial \ell^c$	$\partial \ell^x$	$\partial \ell$	∂q	∂y
$\partial A > 0$	$+$	$+$	0	$+$	$+$	-	$+$

- Consider an aggregate technological shock $\partial A > 0$.
- Things work pretty much like in the RBC model.
- For the c-worker, wealth and substitution effect offset each other in labour supply, so that $\partial \ell^c = 0$. This makes sense as the c-worker has same preferences than the representative agent on the RBC model.
- For the x-worker, preferences are different (only for tractability), so that the substitution effect dominates and ℓ^x increases.
- Therefore, y, c, x and ℓ increase, as in a business cycle.
- As we already said, the problem is that we do not find that A is positively correlated with y in the data.
- Note also that q is countercyclical, which is couterfactual (q goes down because ℓ^x goes up).

c-worker: $\ln c^c - B^c \ell^c + \theta \ln x^c$, x-worker: $\ln(c^x - B^x(\ell^x)^2) + \theta \ln x^x$

	∂c	∂x	$\partial \ell^c$	$\partial \ell^x$	$\partial \ell$	∂q	∂y
$\partial\theta > 0$	$+$	$+$	$+$	$+$	$+$	$+$	$+$

- Consider now a demand shock $\partial\theta > 0$, which is a change in expectations/mood of economic agents, and makes current investment more desirable.
- Contrarily to the RBC model, this demand shock will channel trough the economy because it affects gain from trade between agents.
- Both agents want more investment, the price of investment q goes up and the production of x goes up.
- How can the c-worker get more x^c? By producing more c and trading it against x. There is more trade between the two agents in the economy.
- The increase in the demand for x spills over to the c sector.
- Therefore, both x and c will go up, together with ℓ and y.
- The demand shock creates business cycles-like comovements.

7. Summary

- Business cycles are recurrent fluctuations of economic activity.
- To measure business cycles, one need to eliminate growth and long run movements of economic time series.
- This can be done in the time domain, and the most popular way of doing it is to use the Hodrick & Prescott filter.
- It can also be done in the frequency domain, using spectral analysis.
- Data show that in the business cycle, output, consumption, investment and hours are comoving, are persistent and that productivity is not correlated with output or hours.
- Real Business Cycle models have trouble to explain the comovements in the business cycle.
- One way out is to explicitly model trade linkages within the economy, as done in the Gains From Trade model. (see also Problem set for a “UV” model with sticky prices)

8. References

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